Lecture Notes on Quantum Field Theory II

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1 The Dirac Field

1.1 Lorentz Invariance

Readings

- Peskin & Schroeder Sec. 3.1
- Zee Chap. II

Relativity and Quantum Mechanics

In nonrelativistic quantum mechanics, the dynamics of a particle is described by the Schrdinger equation

$$i\frac{\partial}{\partial t}\psi(t,\mathbf{x}) = -\frac{\nabla^2}{2m}\psi(t,\mathbf{x}). \tag{1.1}$$

However this equation is not compatible with special relativity, because time and space appear differently: the time derivative appears linearly while spatial derivatives appear quadratically.

In special relativity space and time must be treated on equal footing. The fundamental symmetry of spacetime is the **Lorentz symmetry**. **Lorentz invariance** means that if the original fields satisfy the equations of motion, the transformed fields must also satisfy the same equations.

Lorentz Transformations

In special relativity we combine time and space into spacetime coordinates

$$x^\mu = (t, \mathbf{x}) \tag{1.2}$$

where $\mu = 0, 1, 2, 3$. The invariant spacetime interval is

$$s^2 = -t^2 + \mathbf{x}^2. \tag{1.3}$$

Lorentz transformations are linear transformations

$$x'^\mu = \Lambda^\mu{}_\nu x^\nu \tag{1.4}$$

that leave the spacetime interval invariant

$$\eta_{\rho\sigma}\Lambda^\rho{}_\mu\Lambda^\sigma{}_\nu = \eta_{\mu\nu}, \tag{1.5}$$

where the Minkowski metric in our convention is

$$\eta_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{1.6}$$

For example, a boost in the x direction is

$$t' = \gamma(t - vx), \tag{1.7}$$
$$x' = \gamma(x - vt), \tag{1.8}$$

where $\gamma = 1/\sqrt{1-v^2}$.

Relativistic Wave Equations

To construct relativistic wave equations we use Lorentz invariant operators. The derivative operator is

$$\partial_\mu = \left(\frac{\partial}{\partial t}, \nabla\right). \tag{1.9}$$

Using the Minkowski metric we define

$$\partial^\mu = \eta^{\mu\nu}\partial_\nu. \tag{1.10}$$

In our metric convention $(-,+,+,+)$ we obtain

$$\partial_\mu\partial^\mu = -\frac{\partial^2}{\partial t^2} + \nabla^2. \tag{1.11}$$

Using this operator we obtain the Klein-Gordon equation

$$(\partial_\mu\partial^\mu - m^2)\phi = 0. \tag{1.12}$$

This equation is Lorentz invariant and describes a relativistic scalar particle.

Limitations of Relativistic Quantum Mechanics

One may attempt to construct a relativistic quantum theory by writing wave equations for a single particle, such as the Klein-Gordon equation or the Dirac equation. However, this approach faces fundamental difficulties.

- First, in relativistic physics the number of particles is not conserved. Processes such as particle creation and annihilation can occur when the energy is sufficiently large. For example, an electron–positron pair can be created from a photon. A single-particle wave function $\psi(\mathbf{x}, t)$ cannot describe processes in which the number of particles changes.
- Second, the relativistic wave equations introduce negative-energy solutions. Interpreting these states within single-particle quantum mechanics leads to conceptual problems.

These difficulties indicate that relativistic quantum mechanics cannot be consistently formulated as a theory of a fixed number of particles. Instead, the correct framework is quantum field theory, in which fields are promoted to operators defined at each spacetime point and particles are interpreted as excitations of these fields.

We refer the reader to the discussion in Quantum Field Theory I for a more detailed explanation of these issues.

Transformation of Fields

The discussion above indicates that relativistic quantum theory must be formulated in terms of fields rather than single-particle wave functions. A field is a physical quantity defined at every spacetime point,

$$\phi(x), \qquad x^\mu = (t, \mathbf{x}).$$

Since spacetime coordinates transform under Lorentz transformations, fields must transform in a consistent way so that the physical laws remain Lorentz invariant. In the following we discuss how different types of fields transform under Lorentz transformations.

Under a Lorentz transformation the coordinates change as

$$x \to x' = \Lambda\, x. \tag{1.13}$$

Fields transform according to their representation of the Lorentz group.

Scalar field (spin 0) A scalar field is defined as a field whose value is invariant under Lorentz transformations. This means that the value of the field at a given spacetime event does not depend on the choice of inertial frame. For a scalar field we require

$$\phi'(x') = \phi(x). \tag{1.14}$$

This equation states that the value of the field at the same spacetime event is the same in both coordinate systems. Since

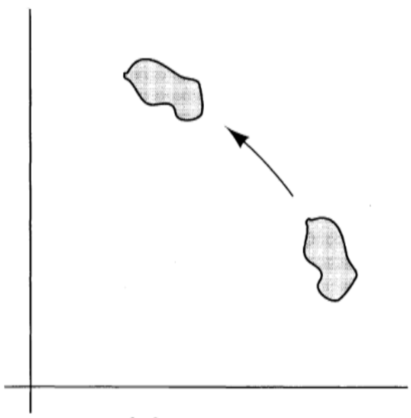
$$x = \Lambda^{-1} x',$$

we can rewrite the transformation law as

$$\phi'(x') = \phi(\Lambda^{-1} x'). \tag{1.15}$$

Finally, renaming the coordinate $x' \to x$, we obtain the commonly used form

$$\boxed{\phi(x) \to \phi'(x) = \phi(\Lambda^{-1} x)} \tag{1.16}$$



(a) scalar field

Example: Lorentz invariance of the Klein–Gordon Lagrangian Consider the Klein–Gordon Lagrangian density

$$\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi + \frac{1}{2}m^2\phi^2. \tag{1.17}$$

Using the scalar field transformation rule

$$\phi \to \phi'(x) = \phi(\Lambda^{-1}x), \tag{1.18}$$

we can determine how the derivatives transform. Using the chain rule we obtain

$$\partial_\mu\phi'(x) = \frac{\partial}{\partial x^\mu}\phi(\Lambda^{-1}x) = (\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\phi(\Lambda^{-1}x). \tag{1.19}$$

Therefore

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = (\Lambda^{-1})^\nu{}_\mu(\Lambda^{-1})^\rho{}_\sigma\eta^{\mu\sigma}\,\partial_\nu\phi(\Lambda^{-1}x)\partial_\rho\phi(\Lambda^{-1}x). \tag{1.20}$$

Using the property of Lorentz transformations

$$\eta_{\mu\nu}\Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}, \tag{1.21}$$

one finds

$$\partial_\mu\phi'(x)\partial^\mu\phi'(x) = \partial_\mu\phi(\Lambda^{-1}x)\partial^\mu\phi(\Lambda^{-1}x). \tag{1.22}$$

Similarly,

$$\phi'^2(x) = \phi^2(\Lambda^{-1}x). \tag{1.23}$$

Therefore the Lagrangian density transforms as

$$\boxed{\mathcal{L}(x) \to \mathcal{L}'(x) = \mathcal{L}(\Lambda^{-1}x)}. \tag{1.24}$$

This means that the Lagrangian density is a Lorentz scalar. The action is

$$S = \int d^4x\,\mathcal{L}(x). \tag{1.25}$$

Since Lorentz transformations satisfy

$$d^4x = d^4x', \tag{1.26}$$

the action is invariant,

$$S' = S. \tag{1.27}$$

Because the equations of motion follow from the stationary condition

$$\delta S = 0, \tag{1.28}$$

the resulting Euler–Lagrange equation

$$(\partial_\mu\partial^\mu - m^2)\phi = 0 \tag{1.29}$$

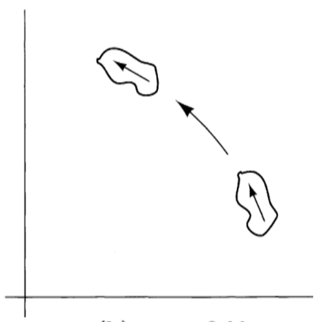
is automatically Lorentz invariant. Therefore, if the Lagrangian density is a Lorentz scalar, the equations of motion derived from it are automatically Lorentz invariant.

Vector field (spin 1) To understand how a vector field transforms, it is useful to first consider ordinary spatial rotations. Let $V^i(x)$ be a vector field in three-dimensional space. Under a rotation R, the coordinates transform as

$$x \to x' = Rx.$$

The vector field transforms as

$$V^i(x) \to V'^i(x) = R^i{}_j\,V^j(R^{-1}x). \tag{1.30}$$



(b) vector field

This transformation has two effects.

- The argument of the field changes from x to $R^{-1}x$ because the spacetime point itself is transformed.
- The components of the vector rotate according to the rotation matrix $R^i{}_j$.

In other words, a rotation affects both the position at which the field is evaluated and the orientation of the vector itself.

The relativistic generalization of this transformation is obtained by replacing rotations with Lorentz transformations. For a four-vector field $V^\mu(x)$ we therefore have

$$\boxed{V^\mu(x) \to V'^\mu(x) = \Lambda^\mu{}_\nu \, V^\nu(\Lambda^{-1}x)}. \tag{1.31}$$

Tensor field The transformation rule for vector fields can be generalized to tensor fields. Consider a rank-two tensor field $T^{\mu\nu}(x)$. Under a Lorentz transformation, the tensor field transforms as

$$\boxed{T^{\mu\nu}(x) \to T'^{\mu\nu}(x) = \Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma T^{\rho\sigma}(\Lambda^{-1}x)}. \tag{1.32}$$

Each Lorentz index introduces one factor of the Lorentz transformation matrix Λ. More generally, for a tensor with multiple indices, each index transforms independently with its own Lorentz matrix.

Index matching and Lorentz covariance Lorentz symmetry imposes a powerful constraint on the form of relativistic equations. If every term in an equation carries the same set of free Lorentz indices, then each term transforms with exactly the same Lorentz matrices. Therefore the equation preserves its form under Lorentz transformations. In this sense, matching Lorentz indices acts as a “machine” that guarantees Lorentz covariance.

E.x. Consider the equation

$$A^\mu + B^\mu = 0. \tag{1.33}$$

Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu{}_\nu A^\nu, \qquad B'^\mu = \Lambda^\mu{}_\nu B^\nu.$$

Both terms acquire the same Lorentz matrix, so the equation becomes

$$\Lambda^\mu{}_\nu A^\nu + \Lambda^\mu{}_\nu B^\nu = 0, \tag{1.34}$$

which is equivalent to the original equation. Therefore the equation is Lorentz covariant.

However, if the indices do not match, the equation will generally not be Lorentz covariant. For example, consider

$$A^\mu + C = 0, \tag{1.35}$$

where C is a scalar. Under a Lorentz transformation we obtain

$$A'^\mu = \Lambda^\mu{}_\nu A^\nu, \qquad C' = C.$$

The two terms now transform differently, so the transformed equation becomes

$$\Lambda^\mu{}_\nu A^\nu + C = 0, \tag{1.36}$$

which is not equivalent to the original equation. Therefore the equation is not Lorentz covariant.

Spinor field (spin-$\frac{1}{2}$) Scalar, vector and tensor fields transform with powers of the Lorentz matrix Λ. However, there exist other fields whose transformation properties cannot be written in this way. These are called **spinor fields**.

A spinor field transforms as

$$\boxed{\psi_a(x) \to \psi'_a(x) = S_{ab}(\Lambda)\, \psi_b(\Lambda^{-1}x)} \tag{1.37}$$

The matrix $S(\Lambda)$ is a representation of the Lorentz group acting on the spinor indices. Unlike vectors, this matrix is not equal to the Lorentz matrix Λ itself. Instead it provides a different representation of the same symmetry.

Continuous symmetry groups The Lorentz symmetry belongs to a class of symmetries called **Lie groups**. A Lie group is a group of transformations that depend smoothly on continuous parameters.

Examples include

- rotations (parameter: angle θ)
- Lorentz transformations (parameters: rotation angles and boosts)
- translations (parameter: displacement a)

Representations A representation of a symmetry group is a way to realize the symmetry transformation using matrices acting on vectors.

For example, a rotation in the plane can be represented by the matrix

$$R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{1.38}$$

Different physical objects transform according to different representations of the same symmetry group.

- scalar representation
- vector representation
- spinor representation

Generators Continuous symmetry transformations can be built from infinitesimal transformations. For example, a small rotation by an angle $d\theta$ can be written as

$$R(d\theta) = I - i\, d\theta\, J. \tag{1.39}$$

The matrix J is called the generator of the rotation.

Then, a finite transformation is obtained by composing infinitely many such infinitesimal transformations. This leads to the exponential form

$$R(\theta) = e^{-i\theta J}. \tag{1.40}$$

Thus, once the generator is known, the full continuous transformation is obtained by exponentiation.

Lie algebra The set of generators together with their commutation relations is called the **Lie algebra** of the group. The commutation relations tell us how different infinitesimal transformations combine.

For example, for spatial rotations the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk} J_k. \tag{1.41}$$

From algebra to representation To construct a representation of the group, we first find matrices that represent the generators and satisfy the same commutation relations. Once such matrices are known, the corresponding finite transformations are obtained by exponentiation.

For example, for spin-$\frac{1}{2}$ particles we may choose

$$J_i = \frac{\sigma_i}{2}, \tag{1.42}$$

where σ_i are the Pauli matrices. Since these matrices satisfy the rotation algebra, they provide a representation of the generators. The corresponding finite rotation is then

$$U(\theta) = e^{-i\theta_i J_i} = e^{-i\theta_i \sigma_i/2}. \tag{1.43}$$

This gives a 2×2 representation of rotations acting on spinors.

Generators of the Lorentz group The Lorentz group is also a Lie group, so its finite transformations are obtained by exponentiating its generators.

To construct the generators, it is useful to first recall the generators of spatial rotations. In three dimensions, the generators of rotations can be written as

$$J_i = i\epsilon_{ijk}x^j\partial_k. \tag{1.44}$$

These generators satisfy the rotation algebra

$$[J_i, J_j] = i\epsilon_{ij}{}^k J_k. \tag{1.45}$$

To generalize this to relativistic spacetime, we introduce antisymmetric generators

$$J^{\mu\nu} = i\left(x^\mu\partial^\nu - x^\nu\partial^\mu\right). \tag{1.46}$$

These are the generators of Lorentz transformations. Because the indices are antisymmetric,

$$J^{\mu\nu} = -J^{\nu\mu}, \tag{1.47}$$

there are six independent generators. They correspond to

- three spatial rotations
- three Lorentz boosts.

Lorentz algebra The generators satisfy the Lorentz Lie algebra

$$[J^{\mu\nu}, J^{\rho\sigma}] = i\left(g^{\nu\rho}J^{\mu\sigma} - g^{\mu\rho}J^{\nu\sigma} - g^{\nu\sigma}J^{\mu\rho} + g^{\mu\sigma}J^{\nu\rho}\right). \tag{1.48}$$

Any matrices that represent Lorentz transformations must satisfy the same commutation relations. Therefore, any representation $M(\Lambda)$ of the Lorentz group must be constructed from matrices that obey this algebra.

Example: vector representation For a four-vector V^μ, the generators are represented by 4×4 matrices

$$(J^{\mu\nu})^\alpha{}_\beta = i\left(\delta^\mu{}_\beta g^{\nu\alpha} - \delta^\nu{}_\beta g^{\mu\alpha}\right). \tag{1.49}$$

An infinitesimal Lorentz transformation can be written as

$$V^\alpha \to \left(\delta^\alpha_\beta - \frac{i}{2}\,\omega_{\mu\nu}(J^{\mu\nu})^\alpha{}_\beta\right)V^\beta, \tag{1.50}$$

where $\omega_{\mu\nu}$ are small antisymmetric parameters.

Examples Different choices of the parameters $\omega_{\mu\nu}$ correspond to different Lorentz transformations.

Rotation example For a rotation in the x-y plane we take

$$\omega_{12} = -\omega_{21} = \theta. \tag{1.51}$$

The corresponding transformation acts on a vector as

$$V \to \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & -\theta & 0 \\ 0 & \theta & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \tag{1.52}$$

Boost example For a boost in the x-direction we take

$$\omega_{01} = -\omega_{10} = \beta. \tag{1.53}$$

The transformation becomes

$$V \to \begin{pmatrix} 1 & \beta & 0 & 0 \\ \beta & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} V. \tag{1.54}$$

Example: spinor representation For a Dirac spinor $\psi_a(x)$, the generators of Lorentz transformations are represented by 4×4 matrices

$$S^{\mu\nu} = \frac{i}{4}\left[\gamma^\mu, \gamma^\nu\right], \tag{1.55}$$

where γ^μ are the Dirac gamma matrices. These matrices satisfy the same Lorentz algebra as the generators $J^{\mu\nu}$ and therefore provide a representation of the Lorentz group acting on spinors. We will introduce more details about the spinor representations in the Dirac fields.

1.2 The Dirac Equation

Readings

- Peskin & Schroeder Sec. 3.2
- Zee Chap. II

Historical motivation

Shortly after the development of quantum mechanics by Schrdinger and Heisenberg, physicists attempted to construct a relativistic wave equation for particles. The first successful relativistic equation was the Klein–Gordon equation,

$$(\partial_\mu \partial^\mu - m^2)\phi = 0, \tag{1.56}$$

which we discussed previously. Although this equation is Lorentz invariant, it is not satisfactory as a single-particle relativistic wave equation.

Two major difficulties arise if the Klein–Gordon equation is interpreted as a wave equation.

- The probability density obtained from the Klein–Gordon current is not positive definite.
- The equation admits negative-energy solutions.

At the time these problems were viewed as technical difficulties rather than indications of a deeper issue. In 1928 Dirac attempted to construct an alternative relativistic wave equation that would overcome these problems.

Dirac's idea was to construct an equation that is

- first order in time derivatives, like the Schrdinger equation,
- Lorentz covariant,
- and reproduces the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \tag{1.57}$$

This led Dirac to propose what is now known as the **Dirac equation**, which provides a relativistic description of spin-$\frac{1}{2}$ particles such as the electron.

Dirac's construction

Dirac began with an equation similar in form to the Schrdinger equation

$$i\partial_t \psi = H\psi, \tag{1.58}$$

where ψ is now a multi-component wavefunction.

Lorentz covariance requires that time and space derivatives appear in a symmetric way. Therefore the Hamiltonian must be linear in spatial derivatives. The most general form is

$$H = \vec{\alpha} \cdot (-i\nabla) + m\beta, \tag{1.59}$$

where $\vec{\alpha}$ and β are constant matrices.

If $\vec{\alpha}$ and β were ordinary numbers, this equation would not be rotationally invariant[1]. Dirac's crucial insight was to interpret α_i and β as matrices acting on a multi-component wave function.

[1] If α_i were numbers, the operator $\alpha_i \partial_i$ would contain a fixed spatial vector $\vec{\alpha}$, which selects a preferred direction and therefore breaks rotational invariance.

Thus the wave function must have several components,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \vdots \\ \psi_n \end{pmatrix}. \tag{1.60}$$

This object is called a **spinor**. Although ψ is written as a column vector, it is not a spacetime vector. The distinction lies in how the object transforms under Lorentz transformations. A vector transforms with the Lorentz matrix $\Lambda^\mu{}_\nu$, while a spinor transforms with a different matrix $S(\Lambda)$ constructed from the generators of the Lorentz group. Objects that transform in this way are called spinors.

A striking difference is that a 360° spatial rotation leaves vectors unchanged but changes the sign of a spinor. Only after a 720° rotation does a spinor return to its original value. This property is characteristic of particles with spin 1/2.

Consistency with relativistic dispersion

To ensure that the equation describes relativistic particles, Dirac required that the equation admit plane-wave solutions with the relativistic dispersion relation

$$p^\mu p_\mu = -m^2. \tag{1.61}$$

A simple way to guarantee this property is to require that squaring the first-order equation reproduces the Klein–Gordon equation. Starting from

$$i\partial_t \psi = (\vec{\alpha} \cdot (-i\nabla) + m\beta)\,\psi, \tag{1.62}$$

we square both sides and obtain

$$-\frac{\partial^2}{\partial t^2}\psi = [\vec{\alpha} \cdot (-i\nabla) + m\beta]^2\,\psi. \tag{1.63}$$

Expanding the right-hand side gives

$$-\alpha^i \alpha^j \partial_i \partial_j \psi + m^2 \beta^2 \psi - im(\beta\alpha^i + \alpha^i \beta)\partial_i \psi. \tag{1.64}$$

For this expression to reduce to the Klein–Gordon equation

$$(-\partial_t^2 + \nabla^2 - m^2)\psi = 0, \tag{1.65}$$

the matrices α_i and β must satisfy the conditions

$$\alpha^i \alpha^j + \alpha^j \alpha^i = 0 \qquad (i \neq j), \tag{1.66}$$

$$(\alpha^i)^2 = 1, \tag{1.67}$$

$$\beta^2 = 1, \tag{1.68}$$

$$\beta\alpha^i + \alpha^i \beta = 0. \tag{1.69}$$

In addition, since the Hamiltonian must be Hermitian, we require

$$\left(\alpha^i\right)^\dagger = \alpha^i, \qquad \beta^\dagger = \beta. \tag{1.70}$$

If these relations are satisfied, equation (1.62) automatically admits plane-wave solutions with the relativistic dispersion relation.

We therefore need to find matrices that satisfy these algebraic conditions. It is easy to check that constant numbers cannot satisfy these relations. One can also verify that 2×2 matrices are not sufficient, and 3×3 matrices also fail to satisfy all the conditions simultaneously. The smallest matrices that work are 4×4 matrices.

Therefore the minimal dimension of the matrices is

$$n \geq 4. \tag{1.71}$$

In fact, there exist infinitely many matrices satisfying these conditions.

Example solutions One possible solution is

$$\beta = \begin{pmatrix} 1_{2\times 2} & 0 \\ 0 & -1_{2\times 2} \end{pmatrix}, \qquad \alpha^i = \begin{pmatrix} 0 & \sigma^i \\ \sigma^i & 0 \end{pmatrix}, \tag{1.72}$$

where σ^i are the Pauli matrices.
Another solution is

$$\beta = \begin{pmatrix} 0 & 1_{2\times 2} \\ 1_{2\times 2} & 0 \end{pmatrix}, \qquad \alpha^i = \begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}. \tag{1.73}$$

One can verify directly that both of these choices satisfy the required relations.

Since α_i and β are 4×4 matrices, the wavefunction ψ must be a four-component object,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix}. \tag{1.74}$$

The components ψ_α are complex functions. This new object is called a **spinor**. Later we will see that it describes particles with spin $\frac{1}{2}$.

Gamma matrices

For later convenience it is useful to rewrite the equation in a more symmetric form. Starting from

$$i\partial_t \psi = -i\vec{\alpha} \cdot \nabla\psi + m\beta\psi, \tag{1.75}$$

we multiply both sides by β,

$$i\beta\partial_t \psi = -i\beta\alpha^i\partial_i\psi + m\psi. \tag{1.76}$$

We now introduce the matrices

$$\gamma^0 = i\beta, \qquad \gamma^i = i\beta\alpha^i. \tag{1.77}$$

In terms of these matrices the equation becomes

$$(\gamma^\mu\partial_\mu - m)\psi = 0. \tag{1.78}$$

Because different authors use different conventions for the Minkowski metric, factors of i may appear in different places in the literature. In our convention the equation takes the form above.

Clifford algebra

The relations satisfied by α_i and β can be written more compactly in terms of the gamma matrices. They satisfy

$$\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 0 \qquad (\mu \neq \nu), \tag{1.79}$$

together with

$$(\gamma^0)^2 = -1, \qquad (\gamma^i)^2 = 1. \tag{1.80}$$

These relations can be written in the compact form

$$\boxed{\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}}. \tag{1.81}$$

Here we use the anticommutator

$$\{A, B\} = AB + BA. \tag{1.82}$$

Equation (1.81) defines the **Clifford algebra**. Any set of matrices γ^μ satisfying this relation is called a representation of the Clifford algebra.

From the above properties of α_i and β we also obtain

$$(\gamma^0)^\dagger = -\gamma^0, \qquad (\gamma^i)^\dagger = \gamma^i. \tag{1.83}$$

These relations can also be written in the compact form

$$\boxed{(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0}. \tag{1.84}$$

Explicit representations of the gamma matrices Using the solutions for α^i and β given earlier, we can construct explicit matrices γ^μ. For example, using the first solution we obtain

$$\gamma^0 = \begin{pmatrix} i\,\mathbf{1}_{2\times 2} & 0 \\ 0 & -i\,\mathbf{1}_{2\times 2} \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix}, \tag{1.85}$$

where σ^i are the Pauli matrices.

Using the second solution for α and β we obtain another set of gamma matrices

$$\gamma^0 = \begin{pmatrix} 0 & i\,\mathbf{1}_{2\times 2} \\ i\,\mathbf{1}_{2\times 2} & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}. \tag{1.86}$$

Both sets of matrices satisfy the Clifford algebra (**??**). Therefore they represent two different representations of the same algebra.

Remarks

- **Massless case**

  Consider the case $m = 0$. In this case the dispersion relation becomes

  $$p^2 = 0. \tag{1.87}$$

  The Dirac equation reduces to

  $$i\partial_t \psi = -i\alpha^i \partial_i \psi. \tag{1.88}$$

  The matrices α_i must satisfy

  $$\{\alpha_i, \alpha_j\} = 0, \qquad \alpha_i^2 = 1. \tag{1.89}$$

  These conditions can be satisfied by the Pauli matrices

  $$\alpha_i = \sigma_i. \tag{1.90}$$

  Therefore in the massless case it is sufficient to use 2×2 matrices, and the wavefunction can be written as a two-component spinor

  $$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}. \tag{1.91}$$

  This illustrates an important difference between massive and massless particles. Massive fermions require four components, while massless fermions can be described by two-component spinors.

  We will discuss this point in more detail later.

- **Conserved current**

  One of Dirac's original motivations was that the Dirac equation admits a conserved current, analogous to the probability current in the Schrdinger equation.

  Starting from the Dirac equation

  $$(\gamma^\mu \partial_\mu - m)\psi = 0, \tag{1.92}$$

  The adjoint equation is

  $$0 = (\partial_\mu \psi^\dagger)\,(\gamma^\mu)^\dagger - m\psi^\dagger = (\partial_\mu \psi^\dagger)\gamma^0\gamma^\mu\gamma^0 - m\psi^\dagger. \tag{1.93}$$

  where we use the relation

  $$(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0. \tag{1.94}$$

  Then, multiplying (1.93) from the right by γ^0 gives

  $$0 = -(\partial_\mu \psi^\dagger)\gamma^0\gamma^\mu - m\psi^\dagger\gamma^0, \tag{1.95}$$

  we define the Dirac adjoint

  $$\bar{\psi} \equiv \psi^\dagger\gamma^0. \tag{1.96}$$

  In the end, one finds that the Hermitian conjugate of the Dirac equation implies the adjoint equation

  $$(\gamma^\mu \partial_\mu + m)\bar{\psi} = 0. \tag{1.97}$$

Multiplying the Dirac equation from the left by $\bar{\psi}$ gives

$$\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi = 0, \tag{1.98}$$

while multiplying the adjoint equation from the right by ψ gives

$$(\partial_\mu\bar{\psi})\gamma^\mu\psi + m\bar{\psi}\psi = 0. \tag{1.99}$$

Adding these two equations, the mass terms cancel and we obtain

$$(\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu\partial_\mu\psi = 0. \tag{1.100}$$

This can be written as

$$\partial_\mu(\bar{\psi}\gamma^\mu\psi) = 0. \tag{1.101}$$

Therefore the conserved current is

$$j^\mu = \bar{\psi}\gamma^\mu\psi. \tag{1.102}$$

In our convention $(\gamma^0)^2 = -1$, so

$$j^0 = -\psi^\dagger\psi. \tag{1.103}$$

Equivalently, one may define

$$\tilde{j}^\mu = -\bar{\psi}\gamma^\mu\psi, \tag{1.104}$$

for which

$$\tilde{j}^0 = \psi^\dagger\psi \geq 0. \tag{1.105}$$

At the end of the day, we have the positive definite conserved current

$$\tilde{j}^\mu = -\bar{\psi}\gamma^\mu\psi \tag{1.106}$$

satisfying $\partial_\mu\tilde{j}^\mu = 0$ with positive definite classically $\tilde{j}^0 = \psi^\dagger\psi \geq 0$.

- **Equivalence of different gamma matrices**

  There are infinitely many matrices satisfying the Clifford algebra. What is the meaning of these different solutions?

  To understand this, consider the Dirac equation as a matrix equation acting on a four-component spinor ψ. Suppose we perform a change of basis in this spinor space

$$\psi \to \psi' = B\psi, \tag{1.107}$$

where B is an invertible constant matrix. Insert $\psi = B^{-1}\psi'$ into the original Dirac equation:

$$(\gamma^\mu\partial_\mu - m)B^{-1}\psi' = 0, \tag{1.108}$$

Multiplying equation (1.78) by B gives

$$(\gamma'^\mu\partial_\mu - m)\psi' = 0, \tag{1.109}$$

where

$$\gamma'^\mu = B\gamma^\mu B^{-1}. \tag{1.110}$$

Thus ψ' satisfies the same Dirac equation but with a different set of gamma matrices.

It is straightforward to check that the matrices γ'^μ also satisfy the Clifford algebra. Therefore any two sets of gamma matrices related by a similarity transformation describe the same physical theory. They correspond simply to different choices of basis for the spinor space.

Although these representations are mathematically equivalent, different choices of gamma matrices can be more convenient in different physical situations. For example, the representation introduced earlier

$$\gamma^0 = \begin{pmatrix} i\mathbf{1}_{2\times2} & 0 \\ 0 & -i\mathbf{1}_{2\times2} \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix} \tag{1.111}$$

is particularly useful for studying the non-relativistic limit, because it makes the connection with the Schrdinger equation more transparent.

On the other hand, another representation

$$\gamma^0 = \begin{pmatrix} 0 & i\mathbf{1}_{2\times2} \\ i\mathbf{1}_{2\times2} & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix} \tag{1.112}$$

is often more convenient in the ultra-relativistic regime.

Therefore different representations of the gamma matrices are used in practice depending on the physical problem being studied. Since all such representations are related by similarity transformations, they lead to the same physical predictions.

1.3 Lorentz covariance of the Dirac equation

Readings

- Peskin & Schroeder Sec. 3.2 and 3.4
- Zee Chap. II

We now show that the Dirac equation is Lorentz covariant. Lorentz covariance means that the equation has the same form in every Lorentz frame. Under a Lorentz transformation,

$$x^\mu \to x'^\mu = \Lambda^\mu{}_\nu x^\nu, \tag{1.113}$$

the transformed field should satisfy the same Dirac equation. Before discussing the Dirac field, let us briefly recall simpler examples.

For a scalar field, the active transformation law is

$$\phi(x) \to \phi'(x) = \phi(\Lambda^{-1}x). \tag{1.114}$$

Similarly, for a vector field such as the Maxwell potential,

$$A_\mu(x) \to A'_\mu(x) = \Lambda_\mu{}^\nu A_\nu(\Lambda^{-1}x). \tag{1.115}$$

In this case the spacetime argument changes and the vector index also transforms with a Lorentz matrix. Maxwell's equations then keep the same form in all frames. For example,

$$\partial_\mu F^{\mu\nu} = 0 \qquad \Longleftrightarrow \qquad \partial'_\mu F'^{\mu\nu} = 0. \tag{1.116}$$

For the Dirac equation, the field ψ is neither a scalar nor a spacetime vector. It is a new object, a spinor. Therefore we must determine how ψ transforms under Lorentz transformations.

Lorentz transformation of the Dirac field

We postulate that under a Lorentz transformation Λ, the Dirac field transforms as

$$\psi(x) \to \psi'(x) = S(\Lambda)\psi(\Lambda^{-1}x), \tag{1.117}$$

where $S(\Lambda)$ is a 4×4 matrix acting on the spinor indices.

This is the active point of view: we keep the coordinate x fixed, but the transformed field at x is expressed in terms of the old field evaluated at the transformed point $\Lambda^{-1}x$.

Lorentz covariance of the Dirac equation means that whenever $\psi(x)$ satisfies

$$(\gamma^\mu\partial_\mu - m)\psi(x) = 0, \tag{1.118}$$

then the transformed field $\psi'(x)$ must also satisfy

$$(\gamma^\mu\partial_\mu - m)\psi'(x) = 0. \tag{1.119}$$

Here the matrices γ^μ themselves do not transform as spacetime vectors. They are fixed matrices acting in spinor space. The Lorentz transformation acts on the field ψ, not on the gamma matrices as dynamical variables.

Condition for Lorentz covariance

To determine $S(\Lambda)$, we compute the derivative of the transformed field:

$$\partial_\mu\psi'(x) = \partial_\mu\left[S(\Lambda)\psi(\Lambda^{-1}x)\right] = S(\Lambda)(\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\psi(\Lambda^{-1}x), \tag{1.120}$$

since $S(\Lambda)$ is a constant matrix for a global Lorentz transformation. Therefore

$$\gamma^\mu\partial_\mu\psi'(x) = \gamma^\mu S(\Lambda)(\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\psi(\Lambda^{-1}x). \tag{1.121}$$

For (1.119) to follow from the original Dirac equation, it is sufficient that

$$\gamma^\mu S(\Lambda)(\Lambda^{-1})^\nu{}_\mu = \gamma^\nu. \tag{1.122}$$

Multiplying this equation from the left by $S^{-1}(\Lambda)$ gives the standard condition

$$S^{-1}(\Lambda)\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\nu\gamma^\nu. \tag{1.123}$$

Equivalently,

$$\boxed{S(\Lambda)\gamma^\mu S^{-1}(\Lambda) = (\Lambda^{-1})^\mu{}_\nu\gamma^\nu}. \tag{1.124}$$

So the problem reduces to finding a matrix $S(\Lambda)$ satisfying (1.123).

Infinitesimal Lorentz transformations

As usual, it is much easier to first solve the problem for infinitesimal Lorentz transformations. Let

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu, \qquad \omega_{\mu\nu} = -\omega_{\nu\mu}, \tag{1.125}$$

where $\omega^\mu{}_\nu$ is infinitesimal. Then

$$(\Lambda^{-1})^\mu{}_\nu = \delta^\mu{}_\nu - \omega^\mu{}_\nu \tag{1.126}$$

to first order in ω. Since Lorentz transformations form a continuous group, the corresponding spinor transformation must also depend smoothly on the parameters $\omega_{\mu\nu}$. Near the identity transformation we can therefore expand $S(\Lambda)$ linearly in $\omega_{\mu\nu}$,

$$S(\Lambda) = 1 - \frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu} + O(\omega^2),$$

where the matrices $\Sigma^{\mu\nu}$ act in spinor space and are called the generators of Lorentz transformations. Similarly,

$$S^{-1}(\Lambda) = 1 + \frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu} \tag{1.127}$$

to first order in ω. Substituting these expressions into (1.123) and keeping only terms linear in ω, we obtain

$$i[\Sigma^{\lambda\rho}, \gamma^\mu] = \eta^{\lambda\mu}\gamma^\rho - \eta^{\rho\mu}\gamma^\lambda. \tag{1.128}$$

Thus the problem is reduced to finding matrices $\Sigma^{\mu\nu}$ satisfying (1.128).

Generators in spinor space

A solution of (1.128) is

$$\Sigma^{\lambda\rho} = \frac{i}{4}[\gamma^\lambda, \gamma^\rho]. \tag{1.129}$$

One can verify this directly by substituting into (1.128) and repeatedly using the Clifford algebra (1.81). The matrices $\Sigma^{\mu\nu}$ are the generators of Lorentz transformations in spinor space. For a finite Lorentz transformation, the corresponding spinor transformation is obtained by exponentiation:

$$S(\Lambda) = \exp\left[-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right]. \tag{1.130}$$

Remarks on the generators

- The matrices Σ^{ij} generate spatial rotations in spinor space:

  $$\Sigma^{ij} = \frac{i}{4}[\gamma^i, \gamma^j]. \tag{1.131}$$

  Since γ^i and γ^j are Hermitian, we find

  $$(\Sigma^{ij})^\dagger = \Sigma^{ij}. \tag{1.132}$$

  Therefore the spinor transformation corresponding to a pure rotation,

  $$S = \exp\left[-\frac{i}{2}\omega_{ij}\Sigma^{ij}\right], \tag{1.133}$$

  is unitary.

- The matrices Σ^{0i} generate boosts in spinor space:

  $$\Sigma^{0i} = \frac{i}{4}[\gamma^0, \gamma^i]. \tag{1.134}$$

  Since $(\gamma^0)^\dagger = -\gamma^0$ and $(\gamma^i)^\dagger = \gamma^i$, we obtain

  $$(\Sigma^{0i})^\dagger = -\Sigma^{0i}. \tag{1.135}$$

  So Σ^{0i} is anti-Hermitian, and the corresponding boost transformation

  $$S = \exp\left[-\frac{i}{2}\omega_{0i}\Sigma^{0i}\right] \tag{1.136}$$

  is not unitary.

- For a general Lorentz transformation containing both rotations and boosts,

  $$SS^\dagger \neq 1. \tag{1.137}$$

  This fact will be important when constructing Lorentz invariant quantities.

Lorentz invariant bilinears

Since S is not unitary in general, the quantity $\psi^\dagger\psi$ is not Lorentz invariant. Indeed,

$$\psi^\dagger\psi \to \psi'^\dagger\psi' = \psi^\dagger(\Lambda^{-1}x)\, S^\dagger S\, \psi(\Lambda^{-1}x), \tag{1.138}$$

which is not equal to $\psi^\dagger\psi$ unless $S^\dagger S = 1$. So we need to find another scalar built from ψ.

From (1.129), one can show that

$$(\Sigma^{\mu\nu})^\dagger = -\gamma^0\Sigma^{\mu\nu}\gamma^0. \tag{1.139}$$

Therefore

$$S^\dagger = \exp\left[\frac{i}{2}\omega_{\mu\nu}(\Sigma^{\mu\nu})^\dagger\right] = \exp\left[-\frac{i}{2}\omega_{\mu\nu}\gamma^0\Sigma^{\mu\nu}\gamma^0\right] = -\gamma^0\exp\left[\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right]\gamma^0. \tag{1.140}$$

Using $(\gamma^0)^2 = -1$, one finds

$$S^\dagger = -\gamma^0 S^{-1}\gamma^0. \tag{1.141}$$

This suggests defining the Dirac adjoint

$$\boxed{\bar\psi \equiv \psi^\dagger\gamma^0}. \tag{1.142}$$

Then

$$\bar\psi\psi \tag{1.143}$$

is a Lorentz scalar. Indeed,

$$\begin{aligned}
\bar\psi'(x)\psi'(x) &= \psi'^\dagger(x)\gamma^0\psi'(x) && (1.144)\\
&= \psi^\dagger(\Lambda^{-1}x)\, S^\dagger\gamma^0 S\, \psi(\Lambda^{-1}x) && (1.145)\\
&= \psi^\dagger(\Lambda^{-1}x)\, \gamma^0\, \psi(\Lambda^{-1}x) && (1.146)\\
&= \bar\psi(\Lambda^{-1}x)\psi(\Lambda^{-1}x). && (1.147)
\end{aligned}$$

It is also useful to write the transformation of $\bar\psi$ itself:

$$\bar\psi(x) \to \bar\psi'(x) = \bar\psi(\Lambda^{-1}x)\, S^{-1}. \tag{1.148}$$

Similarly, from the transformation law of ψ one finds

$$\begin{aligned}
\gamma^\mu\partial_\mu\psi'(x) &= \gamma^\mu S\,(\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\psi(\Lambda^{-1}x) = S\left(S^{-1}\gamma^\mu S\right)(\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\psi(\Lambda^{-1}x)\\
&= S\,\Lambda^\mu{}_\rho\gamma^\rho\,(\Lambda^{-1})^\nu{}_\mu\,\partial_\nu\psi(\Lambda^{-1}x) = S\,\gamma^\mu\partial_\mu\psi(\Lambda^{-1}x).
\end{aligned} \tag{1.149}$$

It is convenient to introduce the slash notation

$$\not{\partial} \equiv \gamma^\mu\partial_\mu. \tag{1.150}$$

Then (1.149) becomes

$$\not{\partial}\,\psi'(x) = S\,\not{\partial}\,\psi(\Lambda^{-1}x). \tag{1.151}$$

One can also show that

$$\bar\psi\gamma^\mu\psi \tag{1.152}$$

transforms as a Lorentz vector.

1.4 Classical solutions of the Dirac Equation

Readings

- Peskin & Schroeder Sec. 3.3
- Zee Chap. II

Dirac action

Using the Lorentz invariant quantities constructed above, we can write the Dirac action as

$$S = -i \int \mathrm{d}^4x \, \bar{\psi} \left(\gamma^\mu \partial_\mu - m \right) \psi. \tag{1.153}$$

This action is Lorentz invariant because $\bar{\psi}\psi$ is a scalar and $\bar{\psi}\gamma^\mu\partial_\mu\psi$ is also a scalar. The overall factor of i is included so that the action is real. Indeed,

$$(\bar{\psi}\psi)^\dagger = -\bar{\psi}\psi \tag{1.154}$$

and

$$(\bar{\psi}\gamma^\mu\partial_\mu\psi)^\dagger = -\bar{\psi}\gamma^\mu\partial_\mu\psi + \text{total derivative}. \tag{1.155}$$

The total derivative contributes only a boundary term to the action, which is assumed to vanish at infinity.

The action (1.153) reproduces the Dirac equation. Treating ψ and $\bar{\psi}$ as independent variables, variation with respect to $\bar{\psi}$ gives

$$(\gamma^\mu\partial_\mu - m)\psi = 0, \tag{1.156}$$

while variation with respect to ψ gives the adjoint equation for $\bar{\psi}$:

$$(\gamma^\mu\partial_\mu + m)\bar{\psi} = 0. \tag{1.157}$$

Classical solutions of the Dirac equation

We have now derived the Dirac equation, determined its Lorentz transformation properties, and constructed the corresponding action. The next step is to study its classical solutions. This is the classical starting point for quantizing the Dirac field.

By construction, the Dirac equation implies the Klein–Gordon equation, so its solutions must be built from plane waves with

$$k^2 = -m^2, \qquad k^\mu = (\omega_{\vec{k}}, \vec{k}). \tag{1.158}$$

However, unlike the scalar field case, this does not completely determine the solution, because the Dirac field has four components.

We therefore separate the solutions into two types:

$$\psi_k^{(+)}(x) = u_k e^{ik\cdot x}, \qquad \psi_k^{(-)}(x) = v_k e^{-ik\cdot x}, \tag{1.159}$$

where u_k and v_k are four-component complex vectors.

The first type is usually called the positive-energy solution, and the second the negative-energy solution. As in the Klein–Gordon case, this terminology refers to the classical plane-wave form. After quantization, the physical excitations all have positive energy.

Substituting these ansätze into the Dirac equation

$$(\gamma^\mu\partial_\mu - m)\psi = 0 \tag{1.160}$$

gives

$$(i\not{k} - m)u_k = 0, \qquad (i\not{k} + m)v_k = 0, \tag{1.161}$$

where

$$\not{k} \equiv k_\mu\gamma^\mu. \tag{1.162}$$

Taking the Dirac adjoint gives

$$\bar{u}_k(i\not{k} - m) = 0, \qquad \bar{v}_k(i\not{k} + m) = 0. \tag{1.163}$$

So the problem reduces to solving these algebraic equations for the spinors u_k and v_k.

Solutions in the rest frame

A convenient shortcut is to first consider the rest frame, where

$$\omega = m, \qquad \vec{k} = 0. \tag{1.164}$$

Then the equations become

$$(-im\gamma^0 - m)u = 0, \qquad (-im\gamma^0 + m)v = 0, \tag{1.165}$$

or equivalently

$$i\gamma^0 u = -u, \qquad i\gamma^0 v = v. \tag{1.166}$$

Thus u and v are eigenvectors of γ^0.

Using the explicit representation

$$\gamma^0 = \begin{pmatrix} iI & 0 \\ 0 & -iI \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & i\sigma^i \\ -i\sigma^i & 0 \end{pmatrix}, \tag{1.167}$$

equation (1.166) implies

$$u^{(0)} = \begin{pmatrix} \xi \\ 0 \end{pmatrix}, \qquad v^{(0)} = \begin{pmatrix} 0 \\ \eta \end{pmatrix}, \tag{1.168}$$

where ξ and η are arbitrary two-component complex vectors.

Choosing a basis, we may take

$$u_1^{(0)} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \qquad u_2^{(0)} = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \tag{1.169}$$

and

$$v_1^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \qquad v_2^{(0)} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}. \tag{1.170}$$

Once the rest-frame solutions are known, one can obtain the solutions for general momentum by performing a Lorentz boost:

$$u_k = Su^{(0)}, \qquad v_k = Sv^{(0)}. \tag{1.171}$$

This is already simpler than solving the original equations directly, although there is an even more efficient method, which we will discuss next.

Constructing solutions for general momentum

Instead of explicitly constructing the Lorentz boost, there is a more efficient method based on a useful algebraic identity.vRecall that

$$\not{k} = \gamma^\mu k_\mu. \tag{1.172}$$

Using the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}, \tag{1.173}$$

one finds

$$\not{k}^2 = \gamma^\mu k_\mu \gamma^\nu k_\nu = k^\mu k_\mu = -m^2, \tag{1.174}$$

where we used the mass-shell condition. From this it follows that

$$(i\not{k} + m)(i\not{k} - m) = -\not{k}^2 - m^2 = 0. \tag{1.175}$$

Using this identity we can construct solutions of the Dirac equation

$$(i\not{k} - m)u = 0, \qquad (i\not{k} + m)v = 0. \tag{1.176}$$

Indeed, if we define [2]

$$u = N(i\not{k} + m)\tilde{u}, \qquad v = N(-i\not{k} + m)\tilde{v}, \tag{1.177}$$

then equation (1.175) guarantees that the Dirac equations are automatically satisfied. Here N is a normalization constant and $\tilde{u}, \tilde{v}$ are arbitrary spinors.

Since we know that there are only two independent solutions for each of u and v, it is convenient to choose the rest-frame solutions as a basis. Thus we take

$$u_s(k) = N(i\not{k} + m)u_s^{(0)}, \qquad v_s(k) = N(-i\not{k} + m)v_s^{(0)}, \qquad s = 1, 2. \tag{1.178}$$

These spinors form a complete basis of solutions to the Dirac equation.

[2] The construction uses a simple algebraic trick. Suppose we want to solve an operator equation $Au = 0$. If there exists another operator B such that $AB = 0$, then any vector of the form

$$u = B\tilde{u}$$

automatically satisfies the equation, since

$$Au = A(B\tilde{u}) = (AB)\tilde{u}.$$

If $AB = 0$, the equation holds for any $\tilde{u}$. In the present case the operators are $A = (i\not{k} - m)$ and $B = (i\not{k} + m)$, and the identity

$$(i\not{k} + m)(i\not{k} - m) = 0$$

plays a role analogous to the elementary factorization $(x + m)(x - m) = 0$. Thus $(i\not{k} + m)$ maps arbitrary spinors into the kernel of $(i\not{k} - m)$, providing solutions of the Dirac equation.

Normalization of the spinors

Taking the Dirac conjugate

$$\bar{u} = u^\dagger \gamma^0, \qquad \bar{v} = v^\dagger \gamma^0, \tag{1.179}$$

one finds[3]

$$\bar{u}_s(k) = N\, \bar{u}_s^{(0)}(i\not{k} + m), \qquad \bar{v}_s(k) = N\, \bar{v}_s^{(0)}(-i\not{k} + m). \tag{1.180}$$

Using the identity (1.175), one immediately obtains

$$\bar{u}_r(k) v_s(k) = 0, \qquad \bar{v}_r(k) u_s(k) = 0. \tag{1.181}$$

We now fix the normalization by requiring

$$\bar{u}_r(k) u_s(k) = i\, 2m\, \delta_{rs}, \qquad \bar{v}_r(k) v_s(k) = -i\, 2m\, \delta_{rs}. \tag{1.182}$$

To determine N, consider

$$\bar{u}_r(k) u_s(k) = N^2\, \bar{u}_r^{(0)}(i\not{k} + m)^2 u_s^{(0)}. \tag{1.183}$$

Using the on-shell relation $\not{k}^2 = -m^2$, one finds

$$(i\not{k} + m)^2 = -\not{k}^2 + 2im\not{k} + m^2 = 2m(i\not{k} + m), \tag{1.184}$$

so that

$$\bar{u}_r(k) u_s(k) = 2mN^2\, \bar{u}_r^{(0)}(i\not{k} + m) u_s^{(0)}. \tag{1.185}$$

Now write

$$\not{k} = \gamma^\mu k_\mu = -E\gamma^0 + k_i \gamma^i, \qquad E = \omega_{\vec{k}}. \tag{1.186}$$

For the rest-frame spinors $u_s^{(0)}$, one has

$$\bar{u}_r^{(0)} u_s^{(0)} = i\, \delta_{rs}, \qquad \bar{u}_r^{(0)} \gamma^0 u_s^{(0)} = -\delta_{rs}, \qquad \bar{u}_r^{(0)} \gamma^i u_s^{(0)} = 0. \tag{1.187}$$

Hence

$$\bar{u}_r^{(0)}(i\not{k} + m) u_s^{(0)} = \bar{u}_r^{(0)}(-iE\gamma^0 + ik_i\gamma^i + m) u_s^{(0)} \tag{1.188}$$

$$= i(E + m)\delta_{rs}. \tag{1.189}$$

Therefore

$$\bar{u}_r(k) u_s(k) = i\, 2m\, N^2 (E + m)\, \delta_{rs}. \tag{1.190}$$

Comparing with the chosen normalization,

$$\bar{u}_r(k) u_s(k) = i\, 2m\, \delta_{rs}, \tag{1.191}$$

we obtain

$$N^2(E + m) = 1. \tag{1.192}$$

Thus the normalization constant is

$$\boxed{N = \frac{1}{\sqrt{E + m}}.} \tag{1.193}$$

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[3]Starting from $u_s(k) = N(i\not{k} + m)u_s^{(0)}$, taking the Hermitian conjugate and multiplying by γ^0 gives

$$\bar{u}_s(k) = u_s^\dagger(k)\gamma^0 = N\big(u_s^{(0)}\big)^\dagger (i\not{k} + m)^\dagger \gamma^0.$$

Using $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$, one obtains

$$(i\not{k} + m)^\dagger \gamma^0 = \gamma^0 (i\not{k} + m),$$

and therefore

$$\bar{u}_s(k) = N\, \bar{u}_s^{(0)}(i\not{k} + m).$$

Similarly,

$$\bar{v}_s(k) = N\, \bar{v}_s^{(0)}(-i\not{k} + m).$$

Explicit form of the spinors

Using the explicit representation of the gamma matrices and the rest frame basis spinors, one obtains

$$\boxed{u_s(k) = \begin{pmatrix} \sqrt{E+m}\,\xi_s \\ \dfrac{\vec{\sigma}\cdot\vec{k}}{\sqrt{E+m}}\xi_s \end{pmatrix}, \qquad v_s(k) = \begin{pmatrix} \dfrac{\vec{\sigma}\cdot\vec{k}}{\sqrt{E+m}}\xi_s \\ \sqrt{E+m}\,\xi_s \end{pmatrix}.} \tag{1.194}$$

The two-component spinors are

$$\xi_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \xi_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{1.195}$$

Non-relativistic limit

In the non-relativistic limit

$$\frac{|\vec{k}|}{m} \ll 1 \Longleftrightarrow E \simeq m, \tag{1.196}$$

the solutions reduce to

$$u_s(k) = \sqrt{2m}\begin{pmatrix} \xi_s \\ 0 \end{pmatrix}, \qquad v_s(k) = \sqrt{2m}\begin{pmatrix} 0 \\ \xi_s \end{pmatrix}. \tag{1.197}$$

Thus the four-component Dirac spinor effectively reduces to the two-component Pauli spinor ξ_s describing a spin-$\frac{1}{2}$ particle in non-relativistic quantum mechanics.

Ultra-relativistic limit

In the ultra-relativistic limit

$$E \gg m, \qquad E \simeq |\vec{k}|, \qquad \hat{k} = \frac{\vec{k}}{|\vec{k}|}, \tag{1.198}$$

the spinors become

$$u_s(k) = \sqrt{E}\begin{pmatrix} \xi_s \\ \vec{\sigma}\cdot\hat{k}\,\xi_s \end{pmatrix}, \qquad v_s(k) = \sqrt{E}\begin{pmatrix} \vec{\sigma}\cdot\hat{k}\,\xi_s \\ \xi_s \end{pmatrix}. \tag{1.199}$$

Using the identity[4]

$$(\vec{\sigma}\cdot\hat{k})^2 = \mathbf{1}_{2\times 2}. \tag{1.200}$$

Since $(\vec{\sigma}\cdot\hat{k})^2 = 1$, defining $\eta_s = (\vec{\sigma}\cdot\hat{k})\xi_s$ implies $\xi_s = (\vec{\sigma}\cdot\hat{k})\eta_s$. Hence

$$v_s(k) = \sqrt{E}\begin{pmatrix} \eta_s \\ \vec{\sigma}\cdot\hat{k}\,\eta_s \end{pmatrix},$$

which has the same form as $u_s(k)$ with ξ_s replaced by η_s. Therefore the v_s spinors lie in the span of u_1, u_2, so only two independent solutions remain in the massless limit. This is consistent with the massless Dirac equation that only requires 2 components.

Useful relations among the spinors

Besides the normalization relations with the Dirac adjoint, the spinors also satisfy the following identities:

$$u_r^\dagger(\vec{k})u_s(\vec{k}) = 2E\,\delta_{rs}, \qquad v_r^\dagger(\vec{k})v_s(\vec{k}) = 2E\,\delta_{rs}. \tag{1.201}$$

Moreover,

$$u_r^\dagger(\vec{k})v_s(-\vec{k}) = 0, \qquad v_r^\dagger(\vec{k})u_s(-\vec{k}) = 0. \tag{1.202}$$

[4] Using $(\vec{\sigma}\cdot\hat{k}) = \sigma^i\hat{k}_i$, we obtain

$$(\vec{\sigma}\cdot\hat{k})^2 = \hat{k}_i\hat{k}_j\sigma^i\sigma^j.$$

Using the Pauli matrix identity

$$\sigma^i\sigma^j = \delta^{ij}I + i\epsilon^{ijk}\sigma^k,$$

this becomes

$$(\vec{\sigma}\cdot\hat{k})^2 = \hat{k}_i\hat{k}_j\delta^{ij}I + i\hat{k}_i\hat{k}_j\epsilon^{ijk}\sigma^k.$$

The first term gives $\hat{k}_i\hat{k}_i = |\hat{k}|^2 = 1$. The second term vanishes because $\hat{k}_i\hat{k}_j$ is symmetric in (i,j) while ϵ^{ijk} is antisymmetric. Hence $(\vec{\sigma}\cdot\hat{k})^2 = I$.

On the other hand,

$$u_r^\dagger(\vec{k})v_s(\vec{k}) \neq 0, \tag{1.203}$$

so the orthogonality here only holds after reversing the spatial momentum.

These relations can be verified directly from the explicit expressions for the spinors. Although the explicit form of $u_s(k)$ and $v_s(k)$ depends on the chosen representation of the gamma matrices, the relations (1.201) and (1.202) are in fact representation-independent. They follow from the Dirac equation and the general algebraic properties of the gamma matrices, and therefore hold in any representation.

Projectors onto positive and negative energy states

One can construct projection operators onto the subspaces spanned by u_s and v_s. The projector onto positive-energy solutions is

$$P_+ = \frac{1}{2im}\sum_{s=1}^{2} u_s(k)\bar{u}_s(k). \tag{1.204}$$

Using the normalization relations one finds

$$P_+^2 = P_+, \tag{1.205}$$

and explicitly

$$P_+ = \frac{i\not{k} + m}{2m}. \tag{1.206}$$

Similarly the projector onto the negative-energy solutions is

$$P_- = \frac{1}{2im}\sum_{s=1}^{2} v_s(k)\bar{v}_s(k) = -\frac{i\not{k} + m}{2m}. \tag{1.207}$$

These projectors satisfy

$$P_-^2 = P_-, \qquad P_+P_- = 0, \qquad P_+ + P_- = 1. \tag{1.208}$$

The operators P_+ and P_- act as projectors onto the subspaces of positive-energy and negative-energy solutions respectively. In particular, one finds

$$P_+u_s(k) = u_s(k), \qquad P_+v_s(k) = 0, \tag{1.209}$$

and

$$P_-v_s(k) = v_s(k), \qquad P_-u_s(k) = 0. \tag{1.210}$$

Thus P_+ projects onto the space spanned by the spinors $u_s(k)$, while P_- projects onto the space spanned by the spinors $v_s(k)$. In this sense the projectors filter out the positive-energy and negative-energy components of a Dirac spinor.

1.5 Quantization of the Dirac field

Readings

- Peskin & Schroeder Sec. 3.5
- Zee Chap. II

Canonical structure

Having obtained a complete set of classical solutions, we now proceed to quantize the Dirac theory. The Dirac action is

$$S = -i\int \mathrm{d}^4x\, \bar{\psi}(\gamma^\mu\partial_\mu - m)\psi. \tag{1.211}$$

To perform canonical quantization we first compute the canonical momentum conjugate to ψ. Treating ψ and $\psi^\dagger$ as independent fields, the Lagrangian density can be written as

$$\mathcal{L} = i\psi^\dagger\partial_0\psi - i\bar{\psi}(\gamma^i\partial_i - m)\psi. \tag{1.212}$$

The canonical momentum is therefore

$$\pi_\psi = \frac{\partial\mathcal{L}}{\partial(\partial_0\psi)} = i\psi^\dagger. \tag{1.213}$$

However,

$$\pi_{\psi^\dagger} = 0, \tag{1.214}$$

which indicates that the Dirac system is a constrained system. The vanishing of $\pi_{\psi^\dagger}$ indicates that the Dirac theory is a constrained system. In general, constrained systems require a more elaborate formalism (such as Dirac's constrained quantization) in order to perform canonical quantization. Fortunately, in the present case there is a simple way to proceed which leads to the correct result.

Since the canonical momentum conjugate to ψ is

$$\pi_\psi = i\psi^\dagger, \tag{1.215}$$

we may interpret ψ as the configuration variable and $\psi^\dagger$ as its canonical momentum. With this interpretation the Lagrangian density can be written in canonical form

$$\mathcal{L} = \pi_\psi\, \partial_0\psi - \mathcal{H}, \tag{1.216}$$

where the Hamiltonian density is

$$\mathcal{H} = i\bar{\psi}(\gamma^i\partial_i - m)\psi. \tag{1.217}$$

Thus the phase space variables are $(\psi, \psi^\dagger)$. We now impose the canonical equal-time commutation relations

$$[\psi_\alpha(t,\vec{x}), \pi_{\psi_\beta}(t,\vec{x}')] = \delta_{\alpha\beta}\,\delta^{(3)}(\vec{x}-\vec{x}'), \qquad \alpha,\beta = 1,2,3,4. \tag{1.218}$$

The next step is to expand the field operator in terms of the complete set of classical solutions,

$$\psi(x) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\vec{k}}}}\sum_{s=1,2}\left[a^{(s)}_{\vec{k}}\, u_s(k)e^{ik\cdot x} + b^{(s)}_{\vec{k}}\, v_s(k)e^{-ik\cdot x}\right]. \tag{1.219}$$

where $u_s(k)e^{-ik\cdot x}$ and $v_s(k)e^{ik\cdot x}$ form a complete set of plane-wave solutions of the Dirac equation. The coefficients $a^{(s)}_{\vec{k}}$ and $b^{(s)}_{\vec{k}}$ are promoted to operators upon quantization. Substituting this expansion into the commutation relations (1.218) leads to

$$[a^{(s)}_{\vec{k}}, (a^{(r)}_{\vec{k}'})^\dagger] = [b^{(s)}_{\vec{k}}, (b^{(r)}_{\vec{k}'})^\dagger] = (2\pi)^3\delta_{rs}\delta^{(3)}(\vec{k}-\vec{k}'), \tag{1.220}$$

with all other commutators vanishing. One may then define the vacuum state $|0\rangle$ by

$$a^{(s)}_{\vec{k}}|0\rangle = 0, \qquad b^{(s)}_{\vec{k}}|0\rangle = 0, \tag{1.221}$$

and build the Hilbert space by acting with the creation operators $(a^{(s)}_{\vec{k}})^\dagger$ and $(b^{(s)}_{\vec{k}})^\dagger$.

Why commutators fail

At first sight this procedure appears to work, but it leads to two serious problems.

- First, the commutation relations imply that the quanta created by $a^\dagger$ and $b^\dagger$ obey bosonic statistics, whereas the Dirac field should describe spin-$\frac{1}{2}$ particles which obey fermionic statistics.
- Second, computing the Hamiltonian

  $$H = \int \mathrm{d}^3x\,\mathcal{H} \tag{1.222}$$

  and substituting the expansion (1.219) gives

  $$H = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\omega_{\vec{k}}\sum_{s=1,2}\left[(a^{(s)}_{\vec{k}})^\dagger a^{(s)}_{\vec{k}} - (b^{(s)}_{\vec{k}})^\dagger b^{(s)}_{\vec{k}}\right]. \tag{1.223}$$

  The minus sign implies that states with many b-particles have arbitrarily negative energy, so the Hamiltonian is not bounded from below and the vacuum would be unstable.

Canonical anticommutation relations

Both problems are resolved by replacing the commutation relations with *anticommutation* relations. Instead of the usual commutator, we impose

$$\left\{\psi_\alpha(t,\vec{x}), \pi_\beta(t,\vec{x'})\right\} = i\delta_{\alpha\beta}\delta^{(3)}(\vec{x}-\vec{x'}), \tag{1.224}$$

or equivalently

$$\left\{\psi_\alpha(t,\vec{x}), \psi^\dagger_\beta(t,\vec{x'})\right\} = \delta_{\alpha\beta}\delta^{(3)}(\vec{x}-\vec{x'}). \tag{1.225}$$

These relations are symmetric between ψ and $\psi^\dagger$ because the anticommutator involves a plus sign rather than a minus sign.

This choice is also self-consistent. Evaluated at $\vec{x}=\vec{x'}$, the left-hand side is positive definite and matches the sign of the right-hand side. The sign ultimately traces back to the relation $\pi_\psi = i\psi^\dagger$ appearing in the Lagrangian.

Mode expansion of the Dirac field

As in the scalar case, we expand the field operator in terms of a complete set of classical solutions,

$$\psi(x) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{\vec{k}}}}\sum_{s=1,2}\left(a^{(s)}_{\vec{k}} u_s e^{ik\cdot x} + \left(b^{(s)}_{\vec{k}}\right)^\dagger v_s e^{-ik\cdot x}\right). \tag{1.226}$$

Here $a^{(s)}_{\vec{k}}$ and $b^{(s)}_{\vec{k}}$ are operators which annihilate particles and antiparticles respectively.

The Dirac spinor has four complex components. Correspondingly, the Dirac equation admits four independent plane-wave solutions: $u_s(k)$ and $v_s(k)$ with $s=1,2$. The u_s describe particles, the v_s describe antiparticles, and the index s labels the two spin polarizations. Thus the four solutions correspond to particle/antiparticle together with spin up/down.

Anticommutation relations for creation operators

Substituting the expansion (1.226) into the canonical anticommutation relations leads to

$$\left\{a^{(s)}_{\vec{k}}, \left(a^{(r)}_{\vec{k'}}\right)^\dagger\right\} = \left\{b^{(s)}_{\vec{k}}, \left(b^{(r)}_{\vec{k'}}\right)^\dagger\right\} = (2\pi)^3\delta^{(3)}(\vec{k}-\vec{k'})\delta_{sr}, \tag{1.227}$$

with all other anticommutators vanishing. The vacuum state is defined by

$$a^{(s)}_{\vec{k}}|0\rangle = b^{(s)}_{\vec{k}}|0\rangle = 0. \tag{1.228}$$

Hamiltonian and vacuum energy

Substituting the mode expansion into the Hamiltonian gives

$$H = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\omega_{\vec{k}}\sum_{s=1,2}\left[\left(a^{(s)}_{\vec{k}}\right)^\dagger a^{(s)}_{\vec{k}} + \left(b^{(s)}_{\vec{k}}\right)^\dagger b^{(s)}_{\vec{k}}\right] + E_0. \tag{1.229}$$

The constant E_0 represents the vacuum energy. The Hamiltonian is now positive definite, so it is natural to define the vacuum state $|0\rangle$ as the state of lowest energy,

$$a^{(s)}_{\vec{k}}|0\rangle = b^{(s)}_{\vec{k}}|0\rangle = 0. \tag{1.230}$$

The vacuum energy E_0 is formally divergent, as in the scalar field case, but it can be removed by normal ordering or by redefining the zero of energy.

Fermionic harmonic oscillators

By analogy with the ordinary harmonic oscillator,

$$H = \omega\left(a^\dagger a + \tfrac{1}{2}\right),$$

a fermionic oscillator is defined by operators satisfying anti-commutation relations

$$\{a, a^\dagger\} = 1, \qquad \{a,a\} = \{a^\dagger, a^\dagger\} = 0. \tag{1.231}$$

Using $\{a, a^\dagger\} = 1$, one can write the Hamiltonian in a symmetric form as

$$H = \omega\left(a^\dagger a - \tfrac{1}{2}\right).$$

However, in quantum field theory the constant term is usually dropped, and one writes

$$H = \omega\, a^\dagger a,$$

since the zero-point energy is physically irrelevant.

Pauli exclusion and spectrum A key consequence of the fermionic algebra is that

$$(a^\dagger)^2 = 0,$$

so the oscillator can only be excited once. Acting on the vacuum,

$$a^\dagger|0\rangle$$

creates a single excitation, while

$$(a^\dagger)^2|0\rangle = 0.$$

Thus the only allowed states are

$$|0\rangle, \qquad a^\dagger|0\rangle.$$

Equivalently, the number operator $N = a^\dagger a$ has eigenvalues 0 and 1. This is the quantum mechanical origin of the Pauli exclusion principle.

Application to the Dirac field For the Dirac field, the Hamiltonian decomposes into independent modes labeled by $(\vec{k}, s)$:

$$H = \sum_{\vec{k},s} \omega_k \left(a_{\vec{k}}^{(s)\dagger} a_{\vec{k}}^{(s)} + b_{\vec{k}}^{(s)\dagger} b_{\vec{k}}^{(s)}\right).$$

For each $(\vec{k}, s)$ there are therefore two independent fermionic oscillators, associated with particles $(a_{\vec{k}}^{(s)})$ and antiparticles $(b_{\vec{k}}^{(s)})$.

Single-particle states

We can now define the one-particle states created from the vacuum. The particle states are

$$|\vec{k}, s\rangle = \sqrt{2\omega_{\vec{k}}}\left(a_{\vec{k}}^{(s)}\right)^\dagger |0\rangle, \tag{1.232}$$

while the antiparticle states are

$$|\vec{k}, \bar{s}\rangle = \sqrt{2\omega_{\vec{k}}}\left(b_{\vec{k}}^{(s)}\right)^\dagger |0\rangle. \tag{1.233}$$

Each particle has two spin polarizations labeled by $s = 1, 2$, and the four-momentum satisfies the mass-shell condition

$$k^2 = -m^2. \tag{1.234}$$

Momentum operator

The Noether charge associated with spatial translations is the momentum operator (See Appendix B)

$$\vec{P} = \int \mathrm{d}^3\vec{x}\, \psi^\dagger(-i\nabla)\psi. \tag{1.235}$$

Using the mode expansion one finds

$$\vec{P} = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\vec{k} \sum_{s=1,2} \left(a^\dagger a + b^\dagger b\right). \tag{1.236}$$

Therefore the states (1.232) and (1.233) are eigenstates of the momentum operator with eigenvalue $\vec{k}$.

Normalization of one-particle states

The normalization of the particle states is

$$\langle \vec{k}, s | \vec{k}', r \rangle = 2\omega_{\vec{k}} \delta_{rs} \delta^{(3)}(\vec{k} - \vec{k}'). \tag{1.237}$$

Similarly for antiparticles,

$$\langle \vec{k}, \bar{s} | \vec{k}', \bar{r} \rangle = 2\omega_{\vec{k}} \delta_{\bar{s}\bar{r}} \delta^{(3)}(\vec{k} - \vec{k}'). \tag{1.238}$$

Particle and antiparticle states are orthogonal,

$$\langle \vec{k}, s | \vec{k}', \bar{r} \rangle = 0, \tag{1.239}$$

since the operators a and b anticommute.

Spin and statistics

Each excitation of the Dirac field has two independent spin polarizations. In quantum mechanics the number of spin states of a particle with spin s is $2s+1$. Since the Dirac particle has two spin states, it follows that

$$2s + 1 = 2 \quad \Rightarrow \quad s = \tfrac{1}{2}.$$

Thus the Dirac field describes spin-$\frac{1}{2}$ particles.

One can verify this explicitly by constructing the angular momentum operator from the Noether charge associated with Lorentz symmetry, see Appendix A.

More generally, the spin-statistics theorem states that fields with half-integer spin must obey anticommutation relations, while fields with integer spin obey commutation relations. Thus fermionic fields obey Fermi–Dirac statistics and satisfy the Pauli exclusion principle, while bosonic fields obey Bose–Einstein statistics.

Charge

Global $U(1)$ symmetry and conserved current

In the Dirac theory, the action

$$S = -i \int \mathrm{d}^4x \, \bar{\psi} \left(\gamma^\mu \partial_\mu - m \right) \psi \tag{1.240}$$

is invariant under a global phase rotation

$$\psi \to e^{i\alpha} \psi, \qquad \bar{\psi} \to e^{-i\alpha} \bar{\psi}, \tag{1.241}$$

where α is a constant. This defines a global $U(1)$ symmetry.

By Noether's theorem, this symmetry implies the existence of a conserved current

$$\partial_\mu j^\mu = 0. \tag{1.242}$$

The associated current is

$$j^\mu = -\bar{\psi} \gamma^\mu \psi, \qquad j^0 = \psi^\dagger \psi. \tag{1.243}$$

In the original interpretation of the Dirac equation as a wave equation, j^0 was interpreted as a probability density and is manifestly positive definite. However, in quantum field theory, ψ is an operator-valued field, and j^μ is interpreted instead as a conserved charge current.

Total charge operator

The conserved charge is defined as

$$Q = \int \mathrm{d}^3x \, j^0 = \int \mathrm{d}^3x \, \psi^\dagger \psi. \tag{1.244}$$

Substituting the mode expansion (1.226), one finds

$$Q = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3} \sum_{s=1,2} \left[\left(a_{\vec{k}}^{(s)} \right)^\dagger a_{\vec{k}}^{(s)} - \left(b_{\vec{k}}^{(s)} \right)^\dagger b_{\vec{k}}^{(s)} \right] + Q_0, \tag{1.245}$$

where Q_0 is an infinite constant arising from the vacuum contribution. As in the Hamiltonian, this constant is removed by normal ordering or by defining the vacuum charge to vanish:

$$Q|0\rangle = 0. \tag{1.246}$$

Particles and antiparticles

After removing the vacuum contribution, the charge operator becomes

$$Q = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3} \sum_{s=1,2} \left[\left(a_{\vec{k}}^{(s)}\right)^\dagger a_{\vec{k}}^{(s)} - \left(b_{\vec{k}}^{(s)}\right)^\dagger b_{\vec{k}}^{(s)} \right]. \tag{1.247}$$

This shows that:

- The operators $a^\dagger$ create states with charge $+1$
- The operators $b^\dagger$ create states with charge -1

Indeed, acting on one-particle states,

$$Q|\vec{k}, s\rangle = +|\vec{k}, s\rangle, \qquad Q|\vec{k}, \bar{s}\rangle = -|\vec{k}, \bar{s}\rangle. \tag{1.248}$$

Thus, the two types of excitations created by $a^\dagger$ and $b^\dagger$ have:

- the same mass
- the same spin ($\frac{1}{2}$)
- opposite charge

This is precisely the definition of particle and antiparticle.

Physical interpretation

The conserved charge Q is an abstract quantum number associated with the global $U(1)$ symmetry. Its physical meaning depends on how the Dirac field is coupled to other fields. When coupled to electromagnetism, this $U(1)$ symmetry becomes the electric charge symmetry, and Q is identified (up to a normalization factor) with electric charge.

Historical remark

One of the most remarkable predictions of the Dirac equation is the existence of antiparticles. Although originally motivated by the search for a relativistic wave equation, the theory naturally predicts states with the same mass and spin as the electron but opposite charge.

Dirac initially speculated that these states might correspond to protons, but this interpretation was quickly ruled out due to the mass difference. The correct interpretation is that they represent a new particle, the positron.

This prediction was experimentally confirmed in 1932 by Carl Anderson, who observed a particle in cosmic rays with the same mass as the electron but opposite charge.

Correlation Functions

Motivation

In quantum field theory, the fundamental observables are not the fields themselves but their correlation functions. While in classical field theory one studies solutions of equations of motion, in quantum theory the fields become operators and cannot be measured directly.

Instead, physical information is encoded in expectation values such as

$$\langle 0|\psi(x_1)\bar{\psi}(x_2)\cdots|0\rangle. \tag{1.249}$$

These correlation functions describe how excitations created at one spacetime point propagate to another.

In particular:

- Two-point functions (propagators) describe the propagation of a single particle.
- Time-ordered correlation functions are the basic building blocks of perturbation theory and Feynman diagrams.
- All scattering amplitudes can ultimately be expressed in terms of correlation functions.

Therefore, instead of solving the Dirac equation directly, quantum field theory focuses on computing correlation functions.

$$\boxed{\text{Correlation functions are the basic observables of quantum field theory.}}$$

The Wightman function

We first define the Wightman function, which does not involve time ordering:

$$D_+^{\alpha\beta}(x-y) \equiv \langle 0|\psi_\alpha(x)\bar{\psi}_\beta(y)|0\rangle. \tag{1.250}$$

Substituting the mode expansion (1.226), only the particle part contributes, and we obtain

$$D_+^{\alpha\beta}(x-y) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{2\omega_{\vec{k}}}\sum_{s=1,2} u_s^\alpha(k)\,\bar{u}_s^\beta(k)\,e^{ik\cdot(x-y)}. \tag{1.251}$$

Using the completeness relation for Dirac spinors,

$$\sum_{s=1,2} u_s(k)\bar{u}_s(k) = (i\not{k}+m), \tag{1.252}$$

we find

$$D_+^{\alpha\beta}(x-y) = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{2\omega_{\vec{k}}} i(i\not{k}+m)_{\alpha\beta}e^{ik\cdot(x-y)}. \tag{1.253}$$

Now we use the fact that

$$k_\mu \to i\partial_\mu^x$$

when acting on the exponential. Therefore,

$$D_+^{\alpha\beta}(x-y) = i(\not{\partial}_x+m)_{\alpha\beta}\int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{2\omega_{\vec{k}}}e^{ik\cdot(x-y)}. \tag{1.254}$$

Defining the scalar Wightman function

$$G_+(x-y) = \langle 0|\phi(x)\phi^\dagger(y)|0\rangle = \int \frac{\mathrm{d}^3\vec{k}}{(2\pi)^3}\frac{1}{2\omega_{\vec{k}}}e^{ik\cdot(x-y)}, \tag{1.255}$$

we obtain the important relation

$$\boxed{D_+^{\alpha\beta}(x-y) = i(\not{\partial}_x+m)_{\alpha\beta}\,G_+(x-y)}. \tag{1.256}$$

This shows that fermionic correlation functions are obtained from scalar ones by acting with a Dirac operator.

Retarded function

The retarded Green's function is defined using the anticommutator:

$$D_R^{\alpha\beta}(x-y) = \theta(x^0-y^0)\langle 0|\{\psi_\alpha(x),\bar{\psi}_\beta(y)\}|0\rangle. \tag{1.257}$$

It satisfies

$$D_R^{\alpha\beta}(x-y) = i(\not{\partial}_x+m)_{\alpha\beta}\,G_R(x-y), \tag{1.258}$$

where $G_R(x-y)$ is the retarded Green's function for the scalar field.

Feynman propagator

The most important object for perturbation theory is the time-ordered (Feynman) propagator:

$$D_F^{\alpha\beta}(x-y) = \langle 0|T\,\psi_\alpha(x)\bar{\psi}_\beta(y)|0\rangle. \tag{1.259}$$

For fermions, time ordering includes an extra minus sign:

$$D_F^{\alpha\beta}(x-y) = \begin{cases} \langle 0|\psi_\alpha(x)\bar{\psi}_\beta(y)|0\rangle, & x^0 > y^0, \\ -\langle 0|\bar{\psi}_\beta(y)\psi_\alpha(x)|0\rangle, & x^0 < y^0. \end{cases} \tag{1.260}$$

This minus sign is a direct consequence of the anticommutation relations of fermionic fields.

As before, one finds

$$\boxed{D_F^{\alpha\beta}(x-y) = i(\not{\partial}_x+m)_{\alpha\beta}\,G_F(x-y)} \tag{1.261}$$

where $G_F(x-y)$ is the scalar Feynman propagator.

Momentum space representation

Taking the Fourier transform, we obtain the propagator in momentum space:

$$D_F(k) = i(i\not{k} + m)\frac{-i}{k^2 + m^2 + i\epsilon}. \tag{1.262}$$

This can be understood from the identity

$$(i\not{k} - m)(i\not{k} + m) = (-k^2 - m^2)\mathbf{1}_{4\times 4}. \tag{1.263}$$

Thus the Dirac propagator can be written compactly as

$$\boxed{D_F(k) = \frac{1}{-i\not{k} + m - i\epsilon}} \tag{1.264}$$

which is a 4×4 matrix in spinor space.

Summary

We see that all fermionic propagators can be obtained from scalar propagators via

$$D(x-y) = i(\not{\partial} + m)\, G(x-y). \tag{1.265}$$

This relation reflects the fact that the Dirac equation is essentially a "square root" of the Klein–Gordon equation.

1.6 Chiral Fermions

Readings

- Peskin & Schroeder Sec. 3.2
- Zee Chap. II

We will start a new topic: **chiral fermions**. Recall that under a Lorentz transformation Λ,

$$x' = \Lambda x, \tag{1.266}$$

the Dirac spinor transforms as

$$\psi(x) \;\rightarrow\; \psi'(x') = S(\Lambda)\,\psi(x), \tag{1.267}$$

where

$$S(\Lambda) = \exp\left(-\frac{i}{2}\,\omega_{\mu\nu}\,\Sigma^{\mu\nu}\right), \qquad \Sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]. \tag{1.268}$$

Thus, Lorentz transformations are generated by the commutators of gamma matrices. This structure ensures that the Dirac equation is consistent with Lorentz symmetry.

Motivation: why study chiral fermions? In the construction of the Dirac equation, we introduced a four-component spinor ψ in order to write a relativistic wave equation that is linear in derivatives and consistent with Lorentz symmetry. This naturally leads to the question:

> Is the four-component structure required by Lorentz symmetry itself, or is it a consequence of additional physical requirements?

To answer this, we analyze how spinors transform under Lorentz transformations. We will find that the Dirac spinor representation is **reducible**: it can be decomposed into two independent two-component pieces. Each of these pieces already furnishes a valid representation of the Lorentz group.

This observation leads to the notion of **chiral (Weyl) fermions**, which describe two-component spinors transforming irreducibly under Lorentz transformations.

The four-component Dirac spinor then arises when we combine the two chiral components, which becomes necessary when additional structures such as a mass term or parity symmetry are imposed. We first study the chiral decomposition of the Dirac spinor.

Choice of gamma matrices To make the reduction explicit, we choose a specific representation - the chiral representation:

$$\gamma^0 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}, \tag{1.269}$$

where σ^i are the Pauli matrices. In this representation, the Lorentz generators take a particularly simple form. Using

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu], \tag{1.270}$$

we obtain the **boost generators**

$$\Sigma^{0i} = -\frac{i}{2}\begin{pmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{pmatrix}. \tag{1.271}$$

Similarly, the rotation generator is

$$\Sigma^{ij} = -\frac{1}{2}\epsilon^{ijk}\begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix}. \tag{1.272}$$

These together form the Lorentz generators. One finds that both boost and rotation generators have the form

$$\Sigma^{\mu\nu} = \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix}, \tag{1.273}$$

which is **block diagonal**. Since

$$S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right), \tag{1.274}$$

it also follows the form of the block matrix

$$S(\Lambda) = \begin{pmatrix} S_L & 0 \\ 0 & S_R \end{pmatrix}. \tag{1.275}$$

Therefore, Lorentz transformations do **not mix** the upper and lower components.

Decomposition of the Dirac spinor We write

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}. \tag{1.276}$$

Then:

- ψ_L transforms with S_L,
- ψ_R transforms with S_R,
- they transform independently.

In summary, each of ψ_L and ψ_R:

- forms a valid representation of the Lorentz group,
- can be treated independently.

Thus,

$$\text{we can consistently reduce from 4 components to 2 components.} \tag{1.277}$$

They are:

- ψ_L: left-handed (chiral) fermion
- ψ_R: right-handed (chiral) fermion

This leads to the notion of **Weyl fermions**.

The **key reason** this works is that the representation is **reducible**: the block diagonal structure of $\Sigma^{\mu\nu}$ implies that the Dirac representation decomposes into two independent pieces.

Why two components are enough From the block diagonal form of $S(\Lambda)$, we see that:

- the upper two components and lower two components do not mix,
- each transforms independently under Lorentz transformations.

Therefore, each of ψ_L and ψ_R already has a well-defined Lorentz transformation. This shows that:

Lorentz covariance itself only requires two-component spinors.

The four-component Dirac spinor arises when we combine the two chiralities, which becomes necessary when introducing a mass term or imposing additional symmetries such as parity.

Relation to the massless case In fact, we already encountered this earlier. For a massless fermion ($m = 0$), the Dirac equation reduces effectively to two components, and remains Lorentz covariant. Thus:

- Massless fermions can be described by two-component spinors.
- Lorentz symmetry is compatible with two components.

This suggests that:

It is the mass term that requires the full four-component Dirac spinor.

General gamma matrices: need for a systematic construction So far, the decomposition $\psi = (\psi_L, \psi_R)$ relied on a special representation (1.269), where the generators are manifestly block diagonal. However, for a **general choice of gamma matrices**, this block structure is not obvious.

Since all representations of gamma matrices are equivalent, the decomposition into two components must exist in any representation. The question is how to construct it in general.

The key object is the matrix

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3. \tag{1.278}$$

The factor of i is chosen such that γ^5 is Hermitian:

$$(\gamma^5)^\dagger = \gamma^5. \tag{1.279}$$

One can verify the following important properties:

- Square:
$$(\gamma^5)^2 = 1 \tag{1.280}$$
- Anticommutation with gamma matrices:
$$\{\gamma^5, \gamma^\mu\} = 0 \tag{1.281}$$
- Traceless:
$$\mathrm{Tr}(\gamma^5) = 0 \tag{1.282}$$

Since γ^5 is Hermitian and satisfies $(\gamma^5)^2 = 1$:

- its eigenvalues are ± 1,
- the trace condition implies equal numbers of $+1$ and -1 eigenvalues.

Therefore, the Dirac spinor space decomposes into two two-dimensional eigenspaces [5].

[5] To clarify the meaning of a "two-dimensional eigenspace," consider

$$\gamma^5 = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}.$$

Its eigenvalues are $\{-1, -1, +1, +1\}$. A convenient choice of corresponding eigenvectors is:

- For eigenvalue $+1$:
$$v_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix}, \qquad v_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

Projection operators Thus, we define the projection operators:

$$P_L = \frac{1-\gamma^5}{2}, \qquad P_R = \frac{1+\gamma^5}{2}. \tag{1.283}$$

They satisfy:

$$P_L^2 = P_L, \qquad P_R^2 = P_R, \qquad P_L P_R = 0, \qquad P_L + P_R = 1. \tag{1.284}$$

The projectors $P_L = \frac{1-\gamma^5}{2}$ and $P_R = \frac{1+\gamma^5}{2}$ project onto the eigenspaces of γ^5 with eigenvalues -1 and $+1$, respectively, i.e.,

$$P_L\gamma^5 = -P_L, \qquad P_R\gamma^5 = +P_R.$$

Definition of chiral components We define:

$$\psi_L = P_L\psi, \qquad \psi_R = P_R\psi. \tag{1.285}$$

These satisfy:

$$\gamma^5\psi_L = -\psi_L, \qquad \gamma^5\psi_R = +\psi_R. \tag{1.286}$$

Thus, ψ_L and ψ_R are eigenstates of γ^5 with eigenvalues -1 and $+1$ respectively. From above, we can see that even for general gamma matrices:

- the Dirac spinor decomposes into two independent components,
- the decomposition is defined by γ^5,
- each component transforms independently under Lorentz symmetry.

This provides a representation-independent definition of **chirality**.

Lorentz invariance of chirality We now show an important property:

$$[\gamma^5, \Sigma^{\mu\nu}] = 0. \tag{1.287}$$

Recall that

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]. \tag{1.288}$$

Using the anticommutation relation

$$\{\gamma^5, \gamma^\mu\} = 0, \tag{1.289}$$

we compute:

$$\begin{aligned}
\gamma^5\Sigma^{\mu\nu} &= \frac{i}{4}\gamma^5(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) && (1.290)\\
&= \frac{i}{4}(-\gamma^\mu\gamma^5\gamma^\nu + \gamma^\nu\gamma^5\gamma^\mu) && (1.291)\\
&= \frac{i}{4}(\gamma^\mu\gamma^\nu\gamma^5 - \gamma^\nu\gamma^\mu\gamma^5) && (1.292)\\
&= \Sigma^{\mu\nu}\gamma^5. && (1.293)
\end{aligned}$$

Thus,

$$[\gamma^5, \Sigma^{\mu\nu}] = 0. \tag{1.294}$$

Since

$$S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right), \tag{1.295}$$

- For eigenvalue -1:

$$w_1 = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}, \qquad w_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \\ 0 \end{pmatrix}.$$

The eigenspace corresponding to eigenvalue $+1$ is defined as all linear combinations of v_1 and v_2:

$$V_+ = \mathrm{span}\{v_1, v_2\}.$$

Similarly, the eigenspace for eigenvalue -1 is

$$V_- = \mathrm{span}\{w_1, w_2\}.$$

Since each eigenspace is generated by two linearly independent vectors, it has dimension 2. In general, the dimension of an eigenspace is the number of independent eigenvectors spanning it.

and γ^5 commutes with all $\Sigma^{\mu\nu}$, it follows that

$$[\gamma^5, S(\Lambda)] = 0. \tag{1.296}$$

Now consider the transformed spinor:

$$\psi'(x') = S(\Lambda)\psi(x). \tag{1.297}$$

Acting with γ^5, we obtain:

$$\gamma^5\psi' = \gamma^5 S(\Lambda)\psi = S(\Lambda)\gamma^5\psi. \tag{1.298}$$

If ψ is an eigenstate of γ^5, for example

$$\gamma^5\psi_L = -\psi_L, \tag{1.299}$$

then

$$\gamma^5\psi'_L = -\psi'_L. \tag{1.300}$$

Similarly,

$$\gamma^5\psi'_R = +\psi'_R. \tag{1.301}$$

Therefore:

- ψ_L transforms into ψ_L,
- ψ_R transforms into ψ_R,
- they do **not mix under Lorentz transformations**.

In Summary, we have shown that:

$$[\gamma^5, S(\Lambda)] = 0 \quad \Rightarrow \quad \text{chirality is preserved under Lorentz transformations.} \tag{1.302}$$

Thus, the decomposition

$$\psi = \psi_L + \psi_R \tag{1.303}$$

is Lorentz invariant, and each component forms a consistent representation of the Lorentz group. This confirms that the notion of **chiral fermions** is well-defined in any representation.

Dirac equation in terms of ψ_L and ψ_R Let us now rewrite the Dirac theory in terms of the chiral components

$$\psi_L = P_L\psi, \qquad \psi_R = P_R\psi. \tag{1.304}$$

In the chiral representation (1.269), the Dirac spinor takes the form

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \tag{1.305}$$

where ψ_L and ψ_R are two-component spinors. Substituting this into the Dirac Lagrangian,

$$\mathcal{L} = -i\bar{\psi}(\gamma^\mu\partial_\mu - m)\psi, \tag{1.306}$$

one can rewrite it in terms of ψ_L and ψ_R. In the chiral representation, the kinetic terms become diagonal, and one finds schematically:

$$\mathcal{L}_{\text{kin}} = i\psi_L^\dagger(\partial_0 + i\sigma^i\partial_i)\psi_L + i\psi_R^\dagger(\partial_0 - i\sigma^i\partial_i)\psi_R. \tag{1.307}$$

Important observation:

- There are **no cross terms** between ψ_L and ψ_R in the kinetic term.

The mass term, however, couples them:

$$\mathcal{L}_{\text{mass}} = -m\left(\psi_L^\dagger\psi_R + \psi_R^\dagger\psi_L\right). \tag{1.308}$$

Now consider the case $m = 0$. Then:

- the mass term vanishes,
- ψ_L and ψ_R completely decouple.

Thus, the theory splits into two independent parts:

$$\mathcal{L} = \mathcal{L}_L + \mathcal{L}_R. \tag{1.309}$$

This confirms again that:

> a massless fermion can be described entirely by a two-component spinor.

This is consistent with our earlier observation that Lorentz symmetry only requires two components.

Chiral symmetry In the massive Dirac theory, we have a global $U(1)$ symmetry:

$$\psi \to e^{i\alpha}\psi. \tag{1.310}$$

In the massless case ($m = 0$), the left- and right-handed components decouple, and can be transformed independently:

$$\psi_L \to e^{i\alpha_L}\psi_L, \qquad \psi_R \to e^{i\alpha_R}\psi_R. \tag{1.311}$$

Therefore, the symmetry is enhanced to:

$$U(1)_L \times U(1)_R. \tag{1.312}$$

This is called a **chiral symmetry**.

Alternative form of chiral symmetry: It is often useful to rewrite the transformation as:

$$\psi \to e^{i\alpha}\psi, \qquad \psi \to e^{i\beta\gamma^5}\psi. \tag{1.313}$$

The second transformation distinguishes ψ_L and ψ_R, since:

$$\gamma^5\psi_L = -\psi_L, \qquad \gamma^5\psi_R = +\psi_R. \tag{1.314}$$

The transformation $e^{i\beta\gamma^5}$ is called an **axial (chiral) symmetry**. It is a symmetry of the theory only in the massless case. When a mass term $m\bar{\psi}\psi$ is present, this symmetry is broken because the mass term mixes ψ_L and ψ_R.

Physical implications Chiral symmetry plays a very important role in physics:

- It is crucial in particle physics (for example, in the physics of pions).
- It can be present at the classical level but may be broken at the quantum level (anomalies).

In particular:

> a classical symmetry may not survive quantization, leading to important physical effects.

Remarks Massless vs massive fermions: the difference between massless and massive fermions is profound:

- Massive fermion:
  - ψ_L and ψ_R are coupled,
  - cannot be separated dynamically.
- Massless fermion:
  - ψ_L and ψ_R are independent,
  - each describes a separate degree of freedom.

Thus:

$$\text{mass is what couples left- and right-handed components.} \tag{1.315}$$

1.7 Majorana Fermions

We now turn to another possible reduction of the Dirac spinor: the **Majorana fermion**.

Motivation: reducing complex degrees of freedom Let us compare different types of fermions:

- Dirac fermion: 4 complex components $\Rightarrow$ 8 real components
- Weyl fermion ψ_L or ψ_R: 2 complex components $\Rightarrow$ 4 real components
- Majorana fermion: 4 real components

Thus, Majorana fermions reduce the degrees of freedom by imposing a **reality condition**.

Basic idea: can a spinor be real? A natural question is:

> Can we impose a reality condition $\psi^* = \psi$ on a Dirac spinor?

At first sight, this seems impossible, because the Dirac equation involves γ^μ, which are generally complex matrices. Let us examine this more carefully.

Compatibility with the Dirac equation The Dirac equation is

$$(\gamma^\mu \partial_\mu - m)\psi = 0. \tag{1.316}$$

Taking complex conjugation gives:

$$(\gamma^{\mu *} \partial_\mu - m)\psi^* = 0. \tag{1.317}$$

If we want $\psi^* = \psi$, then consistency requires:

$$\gamma^{\mu *} = \gamma^\mu. \tag{1.318}$$

Conclusion:

- A real spinor is possible only if we can choose a representation where γ^μ are real matrices.

Majorana representation It turns out that such a representation **does exist**. It is

$$\gamma_M^0 = \begin{pmatrix} 0 & i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix}, \quad \gamma_M^1 = \begin{pmatrix} \sigma^1 & 0 \\ 0 & \sigma^1 \end{pmatrix}, \quad \gamma_M^2 = \begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix} \quad \gamma_M^3 = \begin{pmatrix} \sigma^3 & 0 \\ 0 & \sigma^3 \end{pmatrix} \tag{1.319}$$

In this representation:

$$\gamma_M^\mu = \text{real matrices}, \quad (\gamma_M^\mu)^\dagger = \gamma_M^\mu, \quad (\gamma_M^\mu)^2 = 1 \tag{1.320}$$

Then:

- the Dirac equation is real,
- a real spinor $\psi^* = \psi$ is consistent.

Such a spinor is called a **Majorana spinor**.

Compatibility with Lorentz transformations We must check that the Majorana condition is preserved under Lorentz transformations. Recall that a Dirac spinor transforms as

$$\psi \to \psi' = S(\Lambda)\psi, \qquad S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right). \tag{1.321}$$

In the Majorana representation, the gamma matrices γ_M^μ are real, so their commutators $[\gamma_M^\mu, \gamma_M^\nu]$ are also real. It follows that

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma_M^\mu, \gamma_M^\nu] \tag{1.322}$$

are purely imaginary, and therefore $S(\Lambda)$ is a real matrix. Now suppose that ψ satisfies the Majorana condition

$$\psi^* = \psi. \tag{1.323}$$

After a Lorentz transformation,

$$\psi' = S(\Lambda)\psi, \tag{1.324}$$

we take the complex conjugate:

$$\psi'^* = S(\Lambda)^*\psi^*. \tag{1.325}$$

Since $S(\Lambda)$ is real, $S(\Lambda)^* = S(\Lambda)$, and using $\psi^* = \psi$, we obtain

$$\psi'^* = S(\Lambda)\psi = \psi'. \tag{1.326}$$

Conclusion:

> The Majorana condition is preserved under Lorentz transformations. Therefore, the space of Majorana spinors is closed under Lorentz symmetry.

Relation between different gamma matrix representations So far, the Majorana condition $\psi^* = \psi$ relied on a special representation where γ^μ_M are real. However, in a **general representation**, the gamma matrices are complex, and the condition $\psi^* = \psi$ no longer makes sense. We therefore need a **representation-independent** definition of the Majorana condition.

All sets of gamma matrices satisfying the Clifford algebra are equivalent up to a similarity transformation. In particular, let us denote:

- γ^μ: a general representation,
- γ^μ_M: the Majorana representation (where all γ^μ_M are real).

Then there exists an invertible matrix C such that

$$C\gamma^\mu C^{-1} = \gamma^\mu_M. \tag{1.327}$$

This matrix C implements a **change of basis** from the general representation to the Majorana representation.

Under this change of basis, the spinor transforms accordingly:

$$\psi_M = C\psi. \tag{1.328}$$

In the Majorana representation, we can impose the reality condition:

$$\psi^*_M = \psi_M. \tag{1.329}$$

Substituting $\psi_M = C\psi$, we obtain:

$$(C\psi)^* = C\psi. \tag{1.330}$$

This gives

$$C^*\psi^* = C\psi. \tag{1.331}$$

Multiplying both sides by C^{*-1}, we find:

$$\boxed{\psi^* = B\psi, \qquad B \equiv C^{*-1}C}. \tag{1.332}$$

which is the correct generalization of the Majorana condition.

Properties of B We now derive an important property of B. Starting from

$$C\gamma^\mu C^{-1} = \gamma^\mu_M, \tag{1.333}$$

and using that γ^μ_M are real:

$$(\gamma^\mu_M)^* = \gamma^\mu_M, \tag{1.334}$$

we take the complex conjugate:

$$C^*\gamma^{\mu*}(C^{-1})^* = \gamma^\mu_M. \tag{1.335}$$

Comparing with the original equation, we obtain:

$$C^*\gamma^{\mu*}(C^{-1})^* = C\gamma^\mu C^{-1}. \tag{1.336}$$

Rearranging, we find:

$$\gamma^{\mu*} = B\gamma^\mu B^{-1}, \qquad B = C^{*-1}C. \tag{1.337}$$

Thus, B relates γ^μ to its complex conjugate.

Compatibility with Lorentz transformations We now check that the Majorana condition

$$\psi^* = B\psi \tag{1.338}$$

is preserved under Lorentz transformations. Recall that under a Lorentz transformation:

$$\psi \;\to\; \psi' = S(\Lambda)\psi, \qquad S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right). \tag{1.339}$$

The condition is compatible with Lorentz symmetry if:

$$\boxed{\psi'^* = B\psi'} \tag{1.340}$$

whenever $\psi^* = B\psi$.

Step 1: action of B on $\Sigma^{\mu\nu}$

From

$$\gamma^{\mu *} = B\gamma^{\mu}B^{-1}, \tag{1.341}$$

we derive how B acts on the Lorentz generators:

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma^{\mu}, \gamma^{\nu}]. \tag{1.342}$$

Taking complex conjugation:

$$\Sigma^{\mu\nu *} = \frac{-i}{4}[\gamma^{\mu *}, \gamma^{\nu *}] \tag{1.343}$$

$$= \frac{-i}{4}[B\gamma^{\mu}B^{-1}, B\gamma^{\nu}B^{-1}] \tag{1.344}$$

$$= \frac{-i}{4}B[\gamma^{\mu}, \gamma^{\nu}]B^{-1} \tag{1.345}$$

$$= -B\Sigma^{\mu\nu}B^{-1}. \tag{1.346}$$

Thus:

$$\boxed{\Sigma^{\mu\nu *} = -B\Sigma^{\mu\nu}B^{-1}} \tag{1.347}$$

The minus sign comes from complex conjugation of i.

Step 2: relation for $S(\Lambda)^*$

Now consider the Lorentz transformation matrix:

$$S(\Lambda) = \exp\left(-\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right). \tag{1.348}$$

Taking complex conjugation:

$$S(\Lambda)^* = \exp\left(+\frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu *}\right) \tag{1.349}$$

$$= \exp\left(+\frac{i}{2}\omega_{\mu\nu}(-B\Sigma^{\mu\nu}B^{-1})\right) \tag{1.350}$$

$$= \exp\left(-\frac{i}{2}\omega_{\mu\nu}B\Sigma^{\mu\nu}B^{-1}\right). \tag{1.351}$$

Using the identity

$$\exp(BAB^{-1}) = B\exp(A)B^{-1}, \tag{1.352}$$

we obtain:

$$\boxed{S(\Lambda)^* = B\,S(\Lambda)\,B^{-1}} \tag{1.353}$$

This is a crucial relation.

Step 3: check the Majorana condition

Now compute:

$$\psi'^* = (S(\Lambda)\psi)^* = S(\Lambda)^*\psi^*. \tag{1.354}$$

Using:

$$S(\Lambda)^* = BS(\Lambda)B^{-1}, \qquad \psi^* = B\psi, \tag{1.355}$$

we find:

$$\psi'^* = (BS(\Lambda)B^{-1})(B\psi) \tag{1.356}$$

$$= BS(\Lambda)\psi \tag{1.357}$$

$$= B\psi'. \tag{1.358}$$

Thus:

$$\boxed{\psi'^* = B\psi'} \tag{1.359}$$

We have shown that:

- if ψ satisfies the Majorana condition $\psi^* = B\psi$,
- then the transformed spinor $\psi' = S(\Lambda)\psi$ also satisfies it.

Therefore:

$$\boxed{\text{The Majorana condition is compatible with Lorentz symmetry.}} \tag{1.360}$$

Majorana spinor in the chiral basis Let us now work out the Majorana condition explicitly in the chiral basis. In the Majorana representation, we had simply

$$B = 1. \tag{1.361}$$

In a general basis, however, the Majorana condition takes the form

$$\psi^* = B\psi. \tag{1.362}$$

We now determine B in the chiral representation.

Step 1: reality properties of the gamma matrices. Recall that in the chiral basis,

$$\gamma^0 = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & -i\sigma^i \\ i\sigma^i & 0 \end{pmatrix}. \tag{1.363}$$

From these explicit expressions, one finds:

- γ^0, γ^1, and γ^3 are purely imaginary,
- γ^2 is real.

Thus,

$$(\gamma^0)^* = -\gamma^0, \qquad (\gamma^1)^* = -\gamma^1, \qquad (\gamma^3)^* = -\gamma^3, \tag{1.364}$$

while

$$(\gamma^2)^* = \gamma^2. \tag{1.365}$$

Step 2: determine B. Recall that B is defined by

$$\gamma^{\mu *} = B\gamma^\mu B^{-1}. \tag{1.366}$$

For those gamma matrices that are purely imaginary, this becomes

$$-\gamma^\mu = B\gamma^\mu B^{-1}, \tag{1.367}$$

which means that B must **anticommute** with them. For the gamma matrix that is real, namely γ^2, we have

$$\gamma^2 = B\gamma^2 B^{-1}, \tag{1.368}$$

which means that B must **commute** with γ^2. Therefore, in the chiral basis, B must:

- anticommute with $\gamma^0, \gamma^1, \gamma^3$,
- commute with γ^2.

A matrix with precisely these properties is

$$B = \gamma^2. \tag{1.369}$$

Hence, in the chiral basis, the Majorana condition becomes

$$\boxed{\psi^* = \gamma^2\psi.} \tag{1.370}$$

This is exactly the explicit form of the Majorana condition in the chiral basis.

Step 3: write the condition in terms of ψ_L and ψ_R. Now decompose the Dirac spinor as

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}, \tag{1.371}$$

where ψ_L and ψ_R are two-component spinors. In the chiral basis,

$$\gamma^2 = \begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix}. \tag{1.372}$$

Thus,

$$\psi^* = \begin{pmatrix} \psi_L^* \\ \psi_R^* \end{pmatrix} = \gamma^2\psi = \begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix} \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} = \begin{pmatrix} -i\sigma^2\psi_R \\ i\sigma^2\psi_L \end{pmatrix}. \tag{1.373}$$

Therefore, the Majorana condition implies

$$\psi_L^* = -i\sigma^2\psi_R, \qquad \psi_R^* = i\sigma^2\psi_L. \tag{1.374}$$

Equivalently, we can write

$$\psi_R = i\sigma^2\psi_L^*, \qquad \psi_L = -i\sigma^2\psi_R^*. \tag{1.375}$$

So in the Majorana case, ψ_L and ψ_R are no longer independent. Once one of them is given, the other is completely determined.

Step 4: explicit form of the Majorana spinor. Since ψ_R is determined by ψ_L, a Majorana spinor in the chiral basis can be written as

$$\psi = \begin{pmatrix} \psi_L \\ i\sigma^2 \psi_L^* \end{pmatrix}. \tag{1.376}$$

Thus the whole four-component spinor is determined by a single two-component spinor ψ_L.
This makes the counting of degrees of freedom very clear:

- ψ_L has two complex components,
- therefore it contains four real components,
- so the Majorana spinor has four independent real degrees of freedom.

This agrees with our earlier counting:

$$\text{Majorana fermion} \quad \Longleftrightarrow \quad 4 \text{ real components}. \tag{1.377}$$

Interpretation. In the chiral basis, a Majorana fermion contains both left- and right-handed pieces, but they are related by complex conjugation rather than being independent fields. In this sense, a Majorana spinor is different from a Weyl spinor:

- a Weyl spinor keeps only ψ_L or only ψ_R,
- a Majorana spinor contains both ψ_L and ψ_R, but ties them together by the Majorana condition.

Therefore, the Majorana condition does **not** mean that one simply sets $\psi_R = 0$ or $\psi_L = 0$. Instead, it means that the two chiral components are related to each other.

Remark on the massless case. This condition is independent of whether the fermion is massive or massless. However, in the massless case one may work directly with a single chiral component, and then the description reduces to the same number of degrees of freedom as for a massless Weyl fermion.

A Angular momentum and spin of the Dirac field

The Dirac action is invariant under Lorentz transformations. According to Noether's theorem, this symmetry leads to a conserved current associated with Lorentz transformations. The corresponding conserved charges are the generators of spacetime rotations and boosts.

For an infinitesimal Lorentz transformation

$$x^\mu \to x'^\mu = x^\mu + \omega^\mu{}_\nu x^\nu, \tag{A.1}$$

with antisymmetric parameters $\omega_{\mu\nu} = -\omega_{\nu\mu}$, the Dirac field transforms as

$$\psi(x) \to \left(1 - \frac{i}{2}\omega_{\mu\nu}\Sigma^{\mu\nu}\right)\psi(x), \tag{A.2}$$

where

$$\Sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu] \tag{A.3}$$

are the generators of Lorentz transformations acting on spinors.

The conserved Noether current associated with Lorentz symmetry is

$$M^{\mu\nu\rho} = x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu} - \bar{\psi}\gamma^\mu \Sigma^{\nu\rho}\psi, \tag{A.4}$$

where $T^{\mu\nu}$ is the energy–momentum tensor. The corresponding conserved charges are

$$J^{\nu\rho} = \int d^3x\, M^{0\nu\rho}. \tag{A.5}$$

For spatial indices i, j, these charges generate rotations and therefore represent the angular momentum operator

$$J^{ij} = \int d^3x \left(x^i T^{0j} - x^j T^{0i} - \bar{\psi}\gamma^0 \Sigma^{ij}\psi\right). \tag{A.6}$$

The first two terms correspond to the orbital angular momentum, while the last term represents the intrinsic spin of the Dirac field.

The spin operator is obtained from

$$S^k = \frac{1}{2}\epsilon^{kij}\Sigma^{ij}. \tag{A.7}$$

Using the Dirac matrices one finds

$$\vec{S} = \frac{1}{2}\begin{pmatrix} \vec{\sigma} & 0 \\ 0 & \vec{\sigma} \end{pmatrix}, \tag{A.8}$$

where $\vec{\sigma}$ are the Pauli matrices.

Since the eigenvalues of σ^3 are ± 1, the eigenvalues of the spin operator S_z are

$$S_z = \pm\frac{1}{2}. \tag{A.9}$$

Thus the excitations of the Dirac field carry spin-$\frac{1}{2}$, which explains why each particle has two independent spin polarizations.

B Momentum operator from Noether theorem

In this appendix we derive the expression for the momentum operator of the Dirac field from Noether's theorem.

B.1 Spatial translations

Consider an infinitesimal spacetime translation

$$x^\mu \to x'^\mu = x^\mu + \epsilon^\mu,$$

where ϵ^μ is constant. Under this transformation, the Dirac field transforms as

$$\psi(x) \to \psi'(x) = \psi(x - \epsilon),$$

so that, to first order,

$$\delta\psi(x) = \psi'(x) - \psi(x) = -\epsilon^\mu \partial_\mu \psi(x). \tag{B.1}$$

We focus on spatial translations, $\epsilon^\mu = (0, \vec{\epsilon})$, so that

$$\delta\psi(x) = -\epsilon^i \partial_i \psi(x). \tag{B.2}$$

B.2 Dirac Lagrangian and canonical momentum

The Dirac Lagrangian is

$$\mathcal{L} = -i\bar{\psi}(\gamma^\mu \partial_\mu - m)\psi. \tag{B.3}$$

The derivative of the Lagrangian with respect to $\partial_\mu \psi$ is

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)} = -i\bar{\psi}\gamma^\mu. \tag{B.4}$$

B.3 Noether current and energy-momentum tensor

For spacetime translations, the Noether current is

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \psi)}\,\delta\psi + \mathcal{L}\,\delta x^\mu. \tag{B.5}$$

Since $\delta x^\mu = \epsilon^\mu$ and $\delta\psi = -\epsilon^\nu \partial_\nu \psi$, we obtain

$$j^\mu = -i\bar{\psi}\gamma^\mu(-\epsilon^\nu \partial_\nu \psi) + \mathcal{L}\epsilon^\mu = \epsilon_\nu T^{\mu\nu}, \tag{B.6}$$

where the canonical energy-momentum tensor is

$$T^{\mu\nu} = i\,\bar{\psi}\gamma^\mu \partial^\nu \psi + \eta^{\mu\nu}\mathcal{L}. \tag{B.7}$$

B.4 Momentum operator

The conserved charges associated with translations are

$$P^\nu = \int d^3x\, T^{0\nu}. \tag{B.8}$$

For spatial components, since $\eta^{0i} = 0$ and $(\gamma^0)^2 = -1$,

$$T^{0i} = i\,\bar{\psi}\gamma^0\partial^i\psi = -i\,\psi^\dagger\partial^i\psi. \tag{B.9}$$

Therefore,

$$\boxed{P^i = \int d^3x\, \psi^\dagger(-i\partial^i)\psi.} \tag{B.10}$$

B.5 Interpretation

This expression has the same form as the quantum mechanical momentum operator $-i\nabla$, acting on the field ψ, integrated over all space. It can be interpreted as the total momentum obtained by summing the contribution of each field degree of freedom.